



SCUOLA
ALTI STUDI
LUCCA



Open Lab

A gentle introduction to fMRI: advanced

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A brief outline of the lecture

- **Part A:** bivariate analyses
 - Seed-based Functional Connectivity
 - Functional Connectivity using an Atlas
 - From Functional Connectivity to Graph Analysis
- **Part B:** multivoxel pattern analyses (MVPA)
 - Measuring pattern similarity
 - Decoding approaches
 - Computational models and MVPA
 - Encoding approaches

Two views on the cognitive brain



CS Sherrington
(1857-1952)

- Contemporary cognitive neuroscience, especially animal studies, often takes explanations like sensorimotor *reflexes* as the reference conceptual framework.
- For many cognitive functions, the best model often relies on *computation*, defined as the transformation of *representations* in the brain which result in behaviour.



JJ Hopfield
(1933-)

Two views on the cognitive brain



CS Sherrington
(1857-1952)

- The Sherringtonian view seeks to explain cognition as the result of operations on signals performed at nodes in a network and passed between them. Operations are implemented by specific neurons and their connections in *circuits*.

- The Hopfieldian view explains cognition as the result of transformations of representational spaces that are implemented by *neural populations*. The Hopfieldian view neglects details on connections between specific neurons (or brain regions).



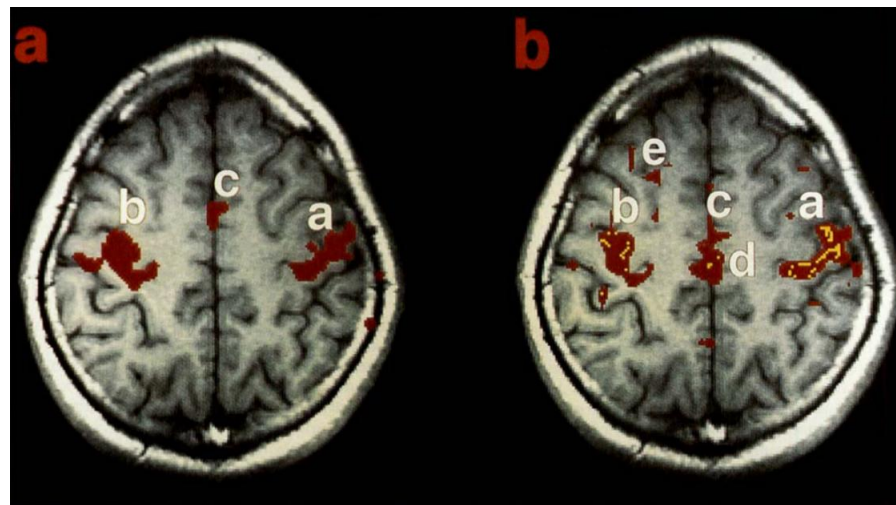
JJ Hopfield
(1933-)

- Besides the differences between the two approaches, each one of them is able to model and account for any neuroimaging experiments.
- The Hopfieldian view is becoming more popular in neuroimaging since it relies on voxels which represent by definition the “average” activity of a neural population, it is able to describe complex cognitive functions, and it accommodates naturalistic stimulations easily.

Part A: from Sherrington to Sporns

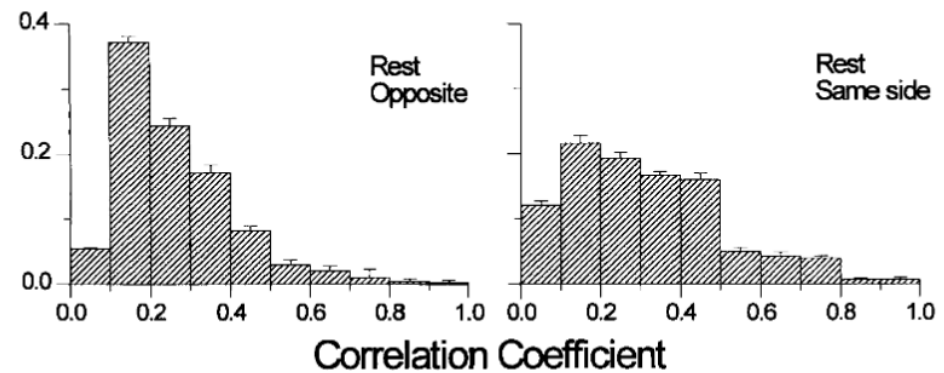
- Some historical notes about Functional Connectivity (FC)
- Strong a-priori: Seed-based FC
- Explorative approach: whole-brain FC using an Atlas
- Building a graph from whole-brain FC

- Functional connectivity is defined as the temporal dependence of neuronal activity patterns of anatomically separated brain regions (Friston, 1993)
- The first paper exploring Functional Connectivity in the human brain motor cortex is from Biswal (1995)

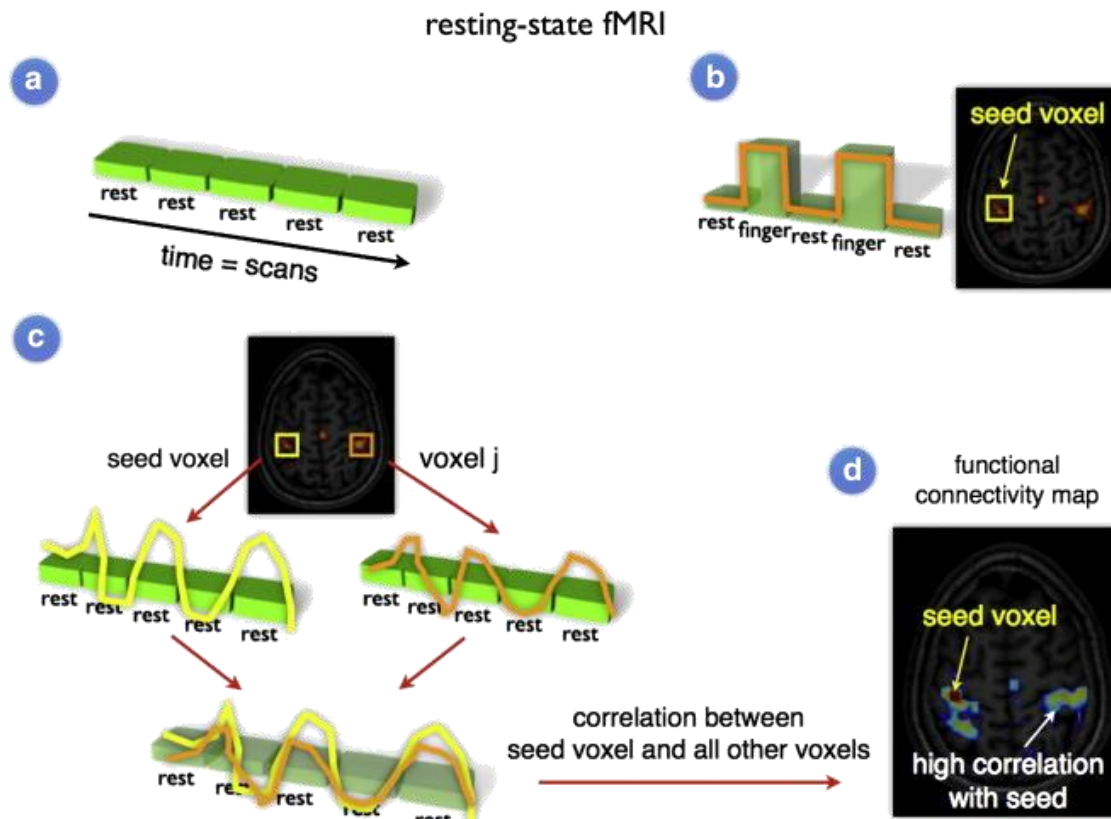


TASK

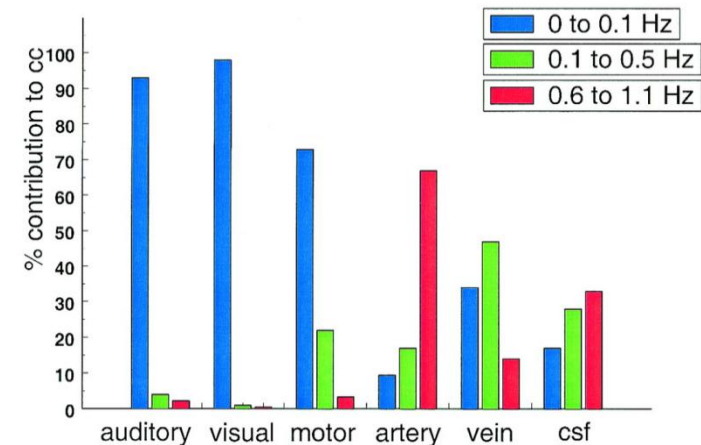
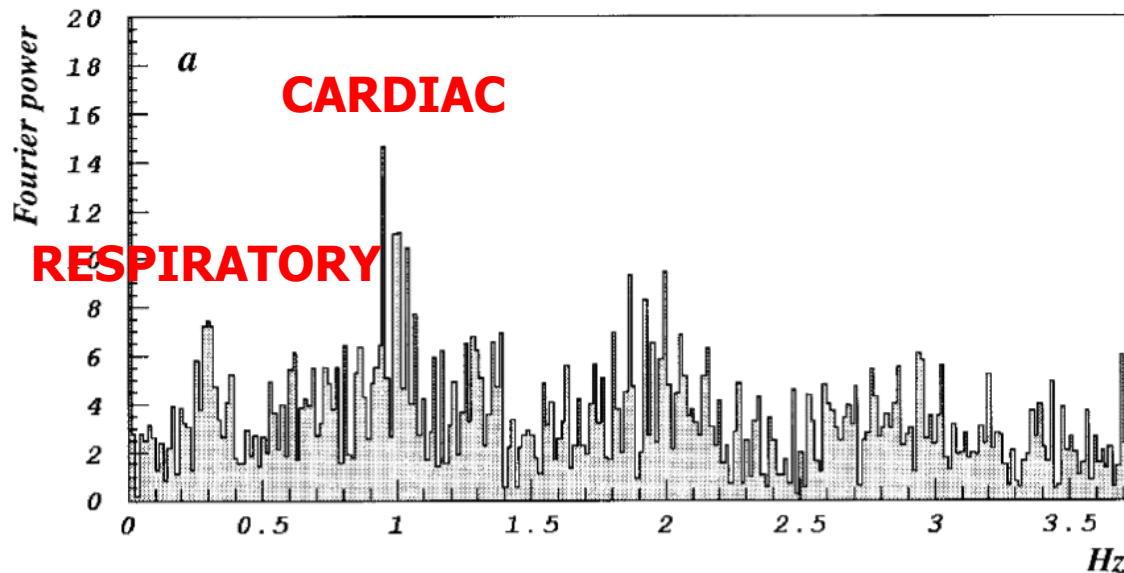
REST



- The evidence that FC may highlight a real functional coupling between regions fostered the field (van den Heuvel, 2010)



- Biswal characterized the frequency range of the functional coupling (<0.1 Hz), which is compatible with the BOLD response
- However the first characterization of possible physiological confounds (beside head movement) was given by Lowe (1998) and Cordes (2000, 2001)



- Subjects acquired in a continuous stable condition (stationary process): resting state or continuous task as naturalistic stimulation
- A clear hypothesis due to the large number of alternative pipelines/methodological choices
- An fMRI dedicated sequence (lowest possible temporal resolution) with possibly the inclusion of physiological confounds (heart rate using finger plethysmography and respiratory rate using a thoracic belt)

Historical Notes on Parcellation

- So far we defined FC as the temporal correlation between a SEED region vs all other voxels of the brain
- Why parcellation of the cerebral cortex is so important?
 - From a coarse (global) perspective, single voxels are redundant: parcellation techniques aim to find discrete neurobiological “atoms”
 - Parcellation reduces voxels from $10^5/10^6$ to $10^2/10^3$!
 - If we want to explore all the possible couplings between each pair of voxel, the overall connectivity matrix will be computationally intractable!

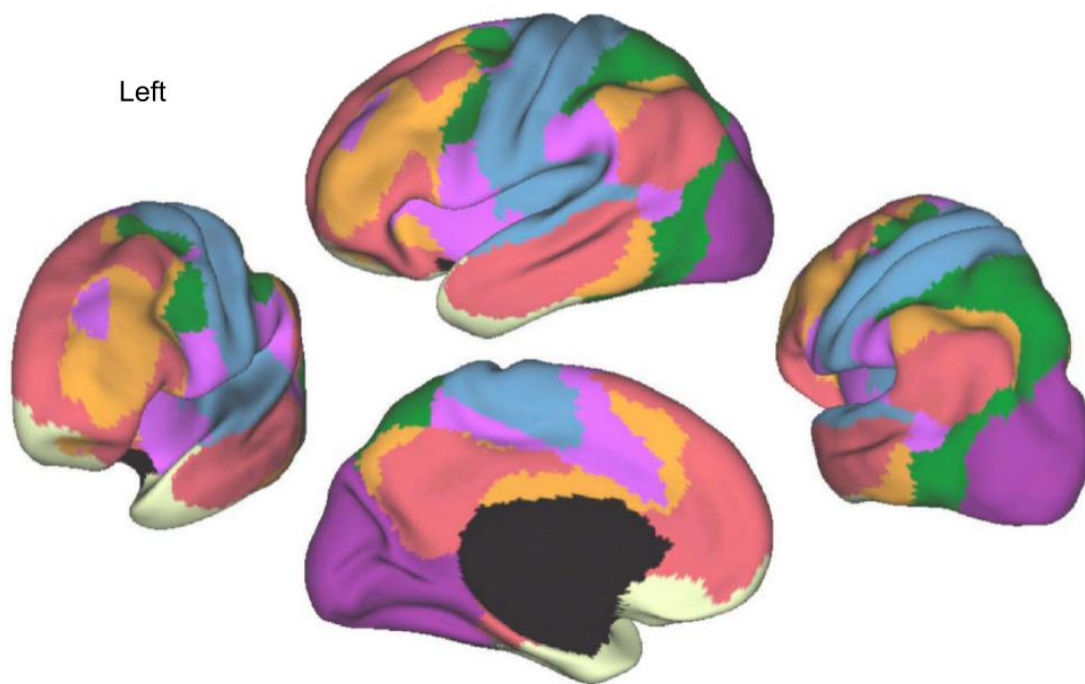
- A plethora of alternatives: <https://www.lead-dbs.org/helpsupport/knowledge-base/atlasresources/cortical-atlas-parcellations-mni-space/>
- Just a brief description of the most useful Atlases (IMHO):
 - Functional Connectivity Atlas 7 Networks (Yeo 2011)
 - Local-Global Parcellation of the Human Cerebral Cortex (Schaefer 2018)
 - HCP MMP 1.0 (Glasser 2016)
 - JuBrain / Juelich histological atlas (Eickhoff, Amunts, Zilles)
 - Harvard-Oxford cortical/subcortical atlases (Desikan 2006)

Historical Notes on Parcellation

Functional Connectivity Atlas

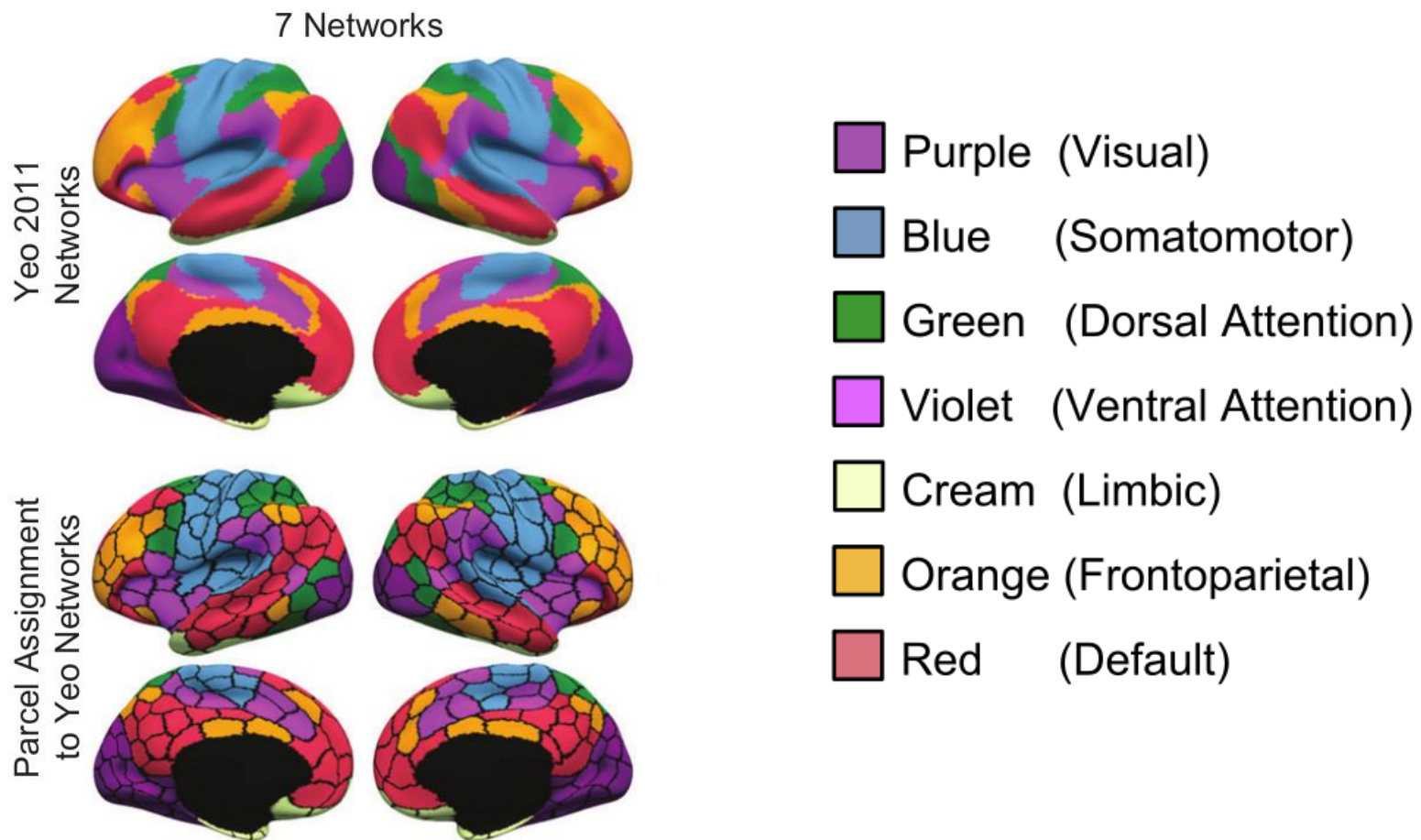
7 Resting State Networks (RSN Yeo 2011)

7-Network Parcellation (N=1000)

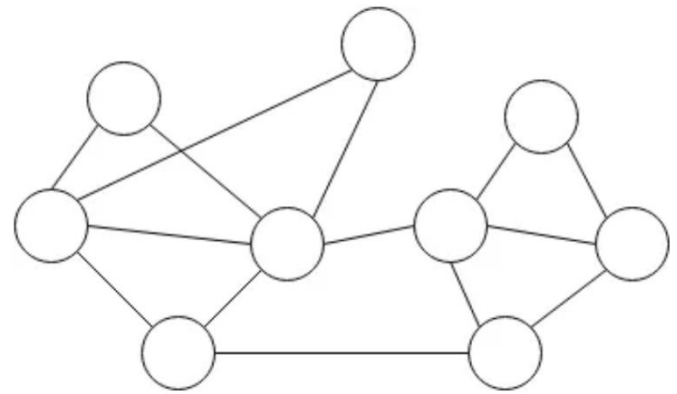


- Purple (Visual)
- Blue (Somatomotor)
- Green (Dorsal Attention)
- Violet (Ventral Attention)
- Cream (Limbic)
- Orange (Frontoparietal)
- Red (Default)

Local-Global Parcellation (Schaefer 2018)



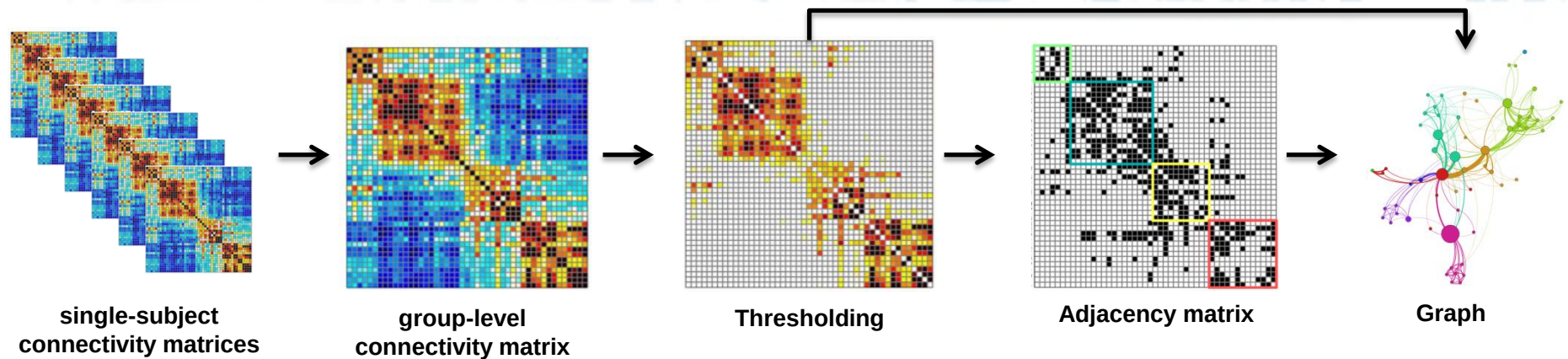
Towards graph analysis



From parcellation to adjacency matrix

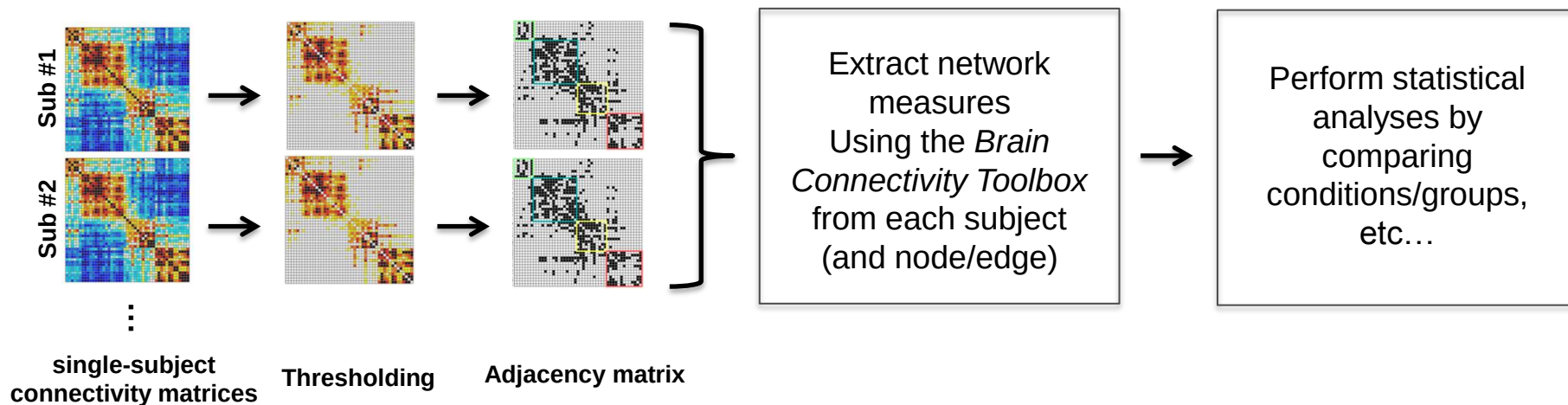
- Generally, after parcellation of functional data (i.e., averaging across voxels, within each ROI), the next step is to evaluate the correlation between each pair of ROIs in each subject to obtain connectivity matrices
- Then, the experimenter has two possibilities: a) build a group-level adjacency matrix and create a graph from it (“fixed-effect” analysis); b) perform a proper statistical analysis (“random-effect” analysis);

From parcellation to adjacency matrix



- The group level analysis allows the creation of a graph that can be used to: 1) represent the data through a meaningful and interpretable graph; 2) perform a statistical comparison between the group-level matrix (or the graph) and some external representations (e.g., a group-level graph created in humans vs one built in monkeys)

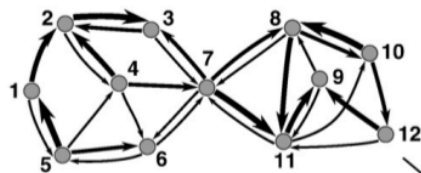
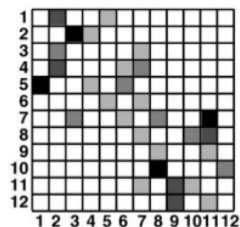
From parcellation to adjacency matrix



- The single-subject analysis to test group-level differences accounting for random-effects

weighted directed networks

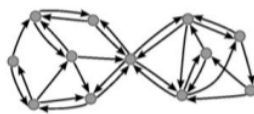
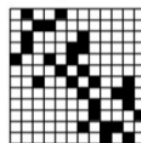
structural datasets: tract tracing
effective datasets: inference of causality
from functional data



binarize

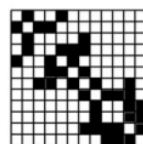
symmetrize

binary directed networks



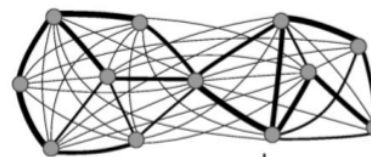
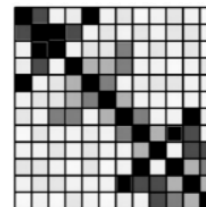
symmetrize

binary undirected networks



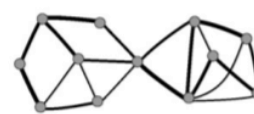
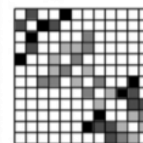
weighted undirected networks

structural datasets: diffusion MRI, structural MRI
functional datasets: functional MRI, MEG, EEG

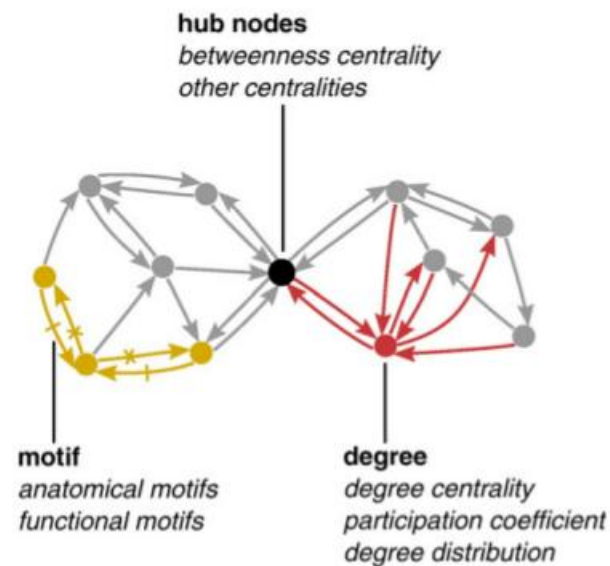
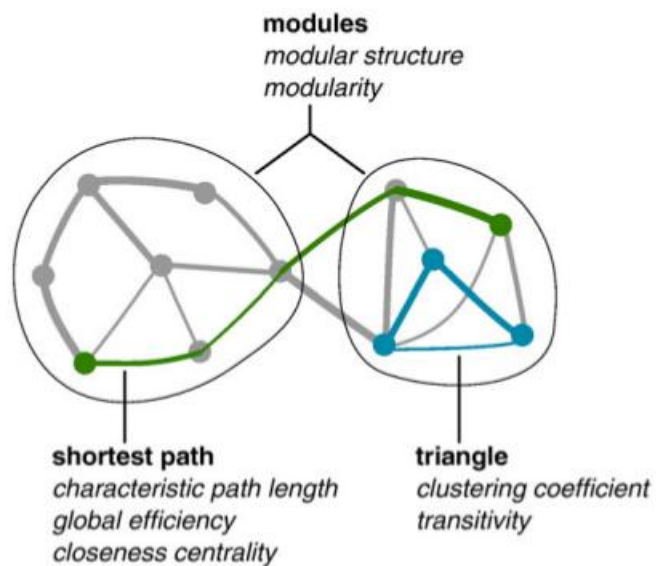


threshold

weighted undirected networks



binarize



- So far we defined FC as an undirected measure of functional coupling, since correlation does not imply causation
- Effective connectivity are a bunch of methods to explore also causal relationship between ROIs (Friston 2011)
- Effective connectivity could generate directed graphs, expanding network analysis

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 - Building feature-based descriptors of stimuli
 - Encoding approaches

- Some historical notes about MVPA
- Advantages and caveats of RSA
- Decoding and classifiers
- Building models from stimulus set
- Encoding approaches

Historical Notes: fMRI perspective

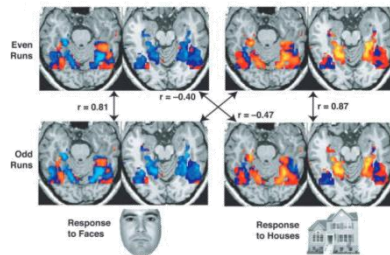


Distributed pattern (Edelman & Malach, 1998)

Historical Notes: fMRI perspective



Distributed pattern (Edelman & Malach, 1998)

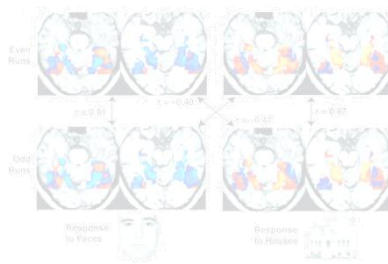


*Overlapping representation of multiple categories
(Haxby & Pietrini, 2001)*

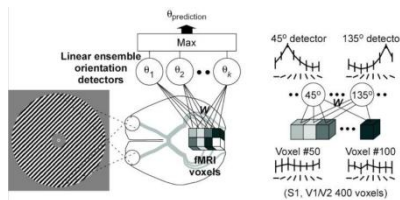
Historical Notes: fMRI perspective



Distributed pattern (Edelman & Malach, 1998)



Overlapping representation of multiple categories (Haxby & Pietrini, 2001)

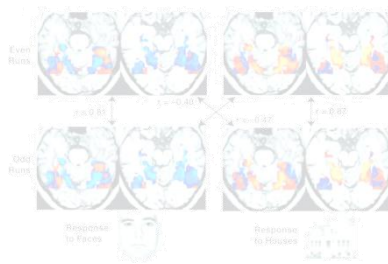


Decoding subjective perception (Kamitani & Tong, 2005)

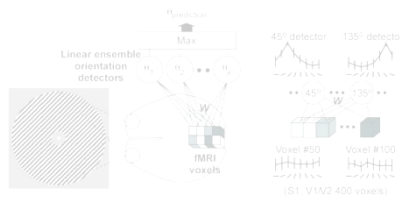
Historical Notes: fMRI perspective



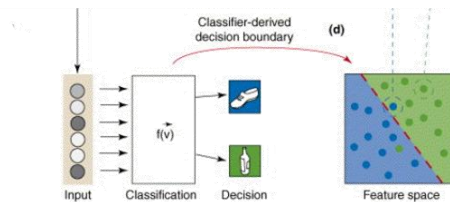
Distributed pattern (Edelman & Malach, 1998)



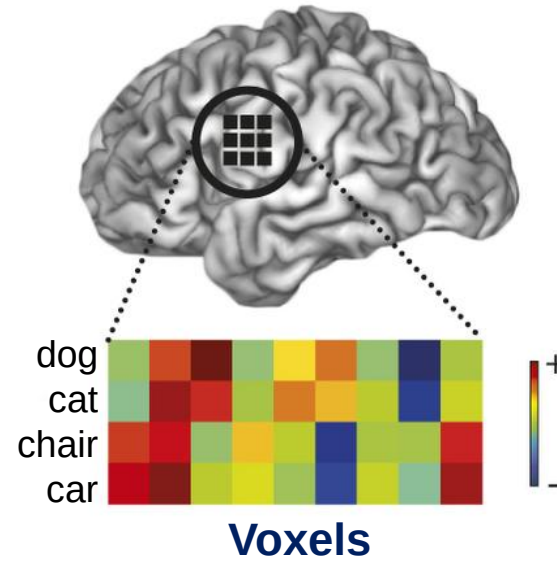
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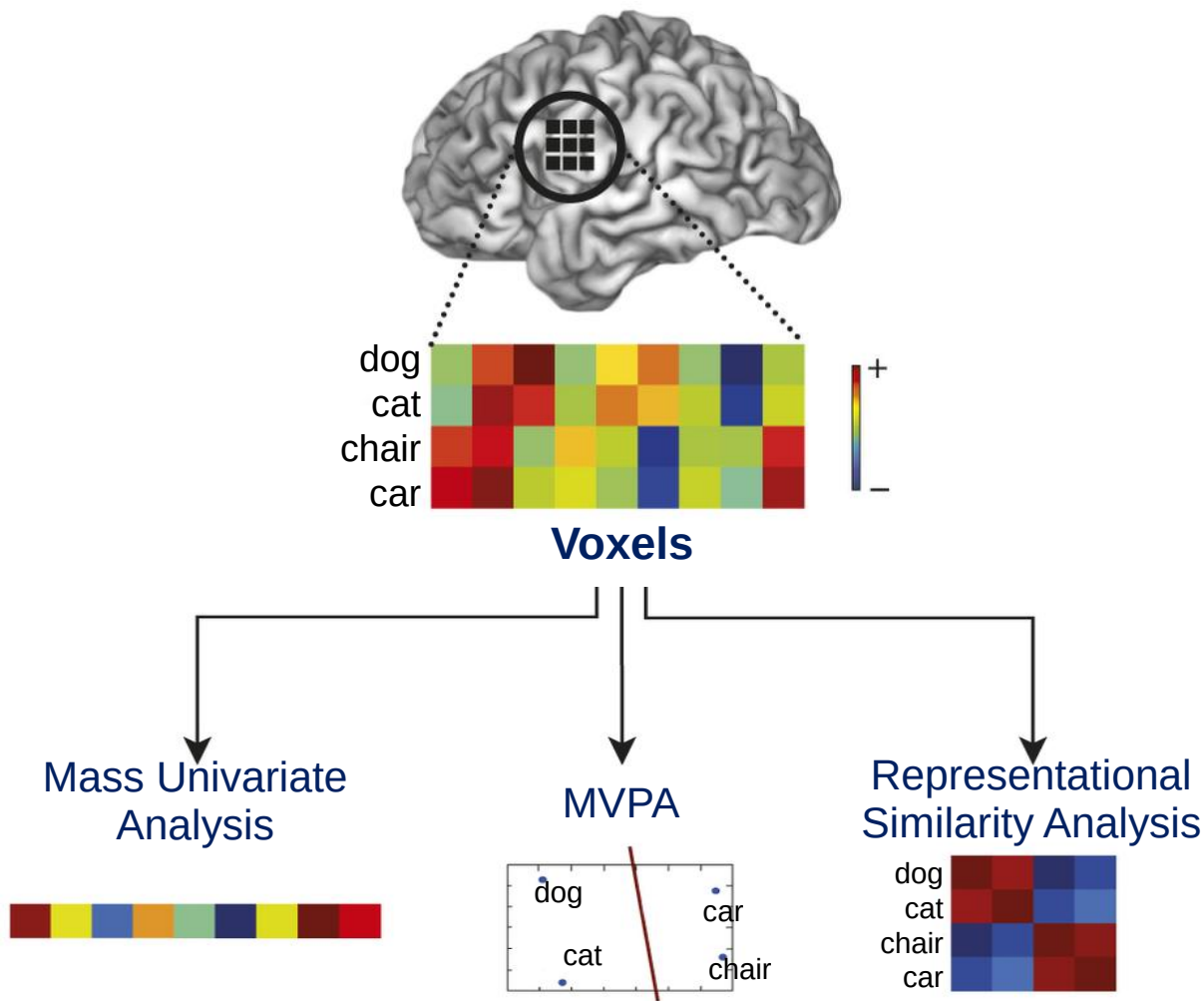


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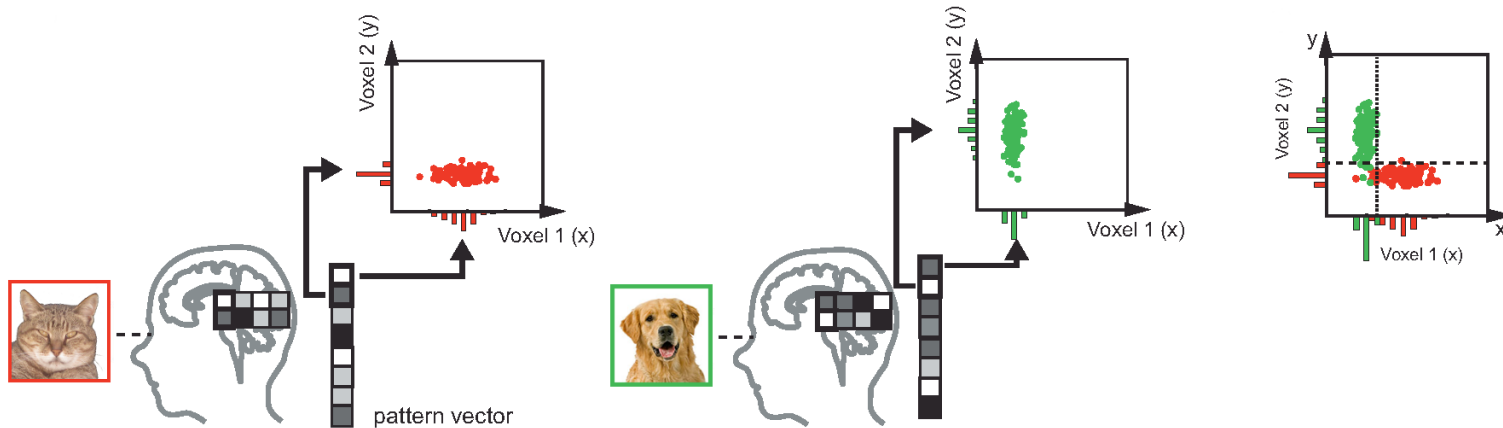


Multi-Voxel Pattern Analysis (MVPA) (Norman & Haxby, 2006)

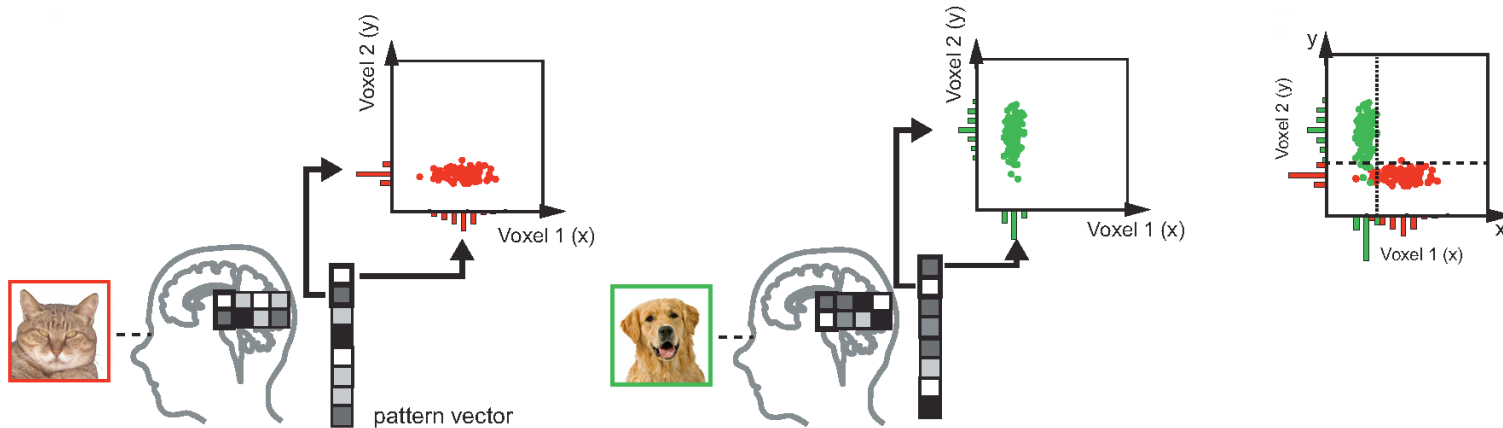




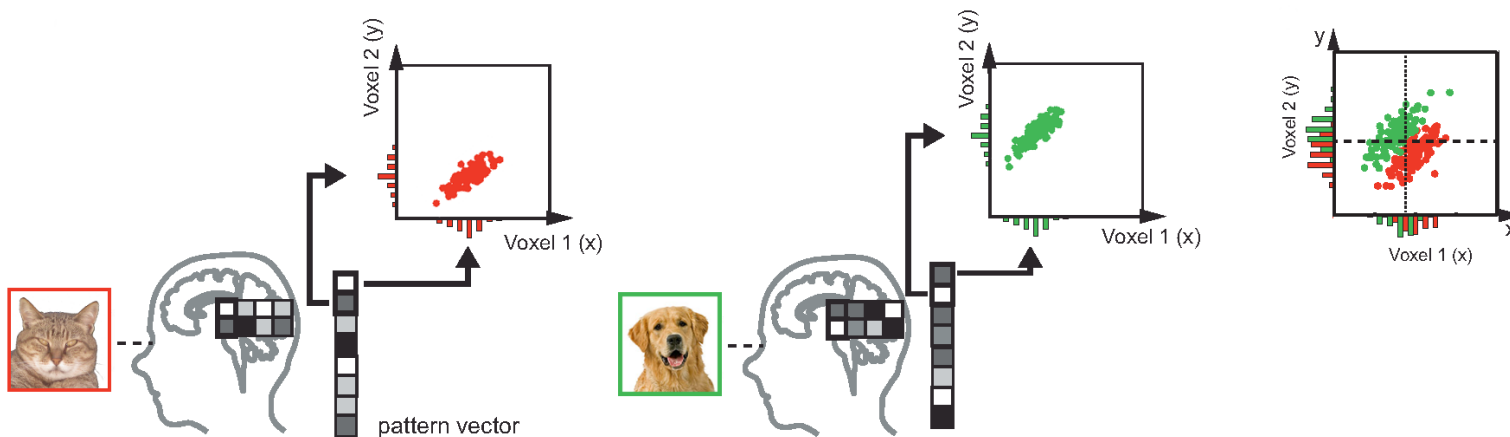
Univariate specific response to categories



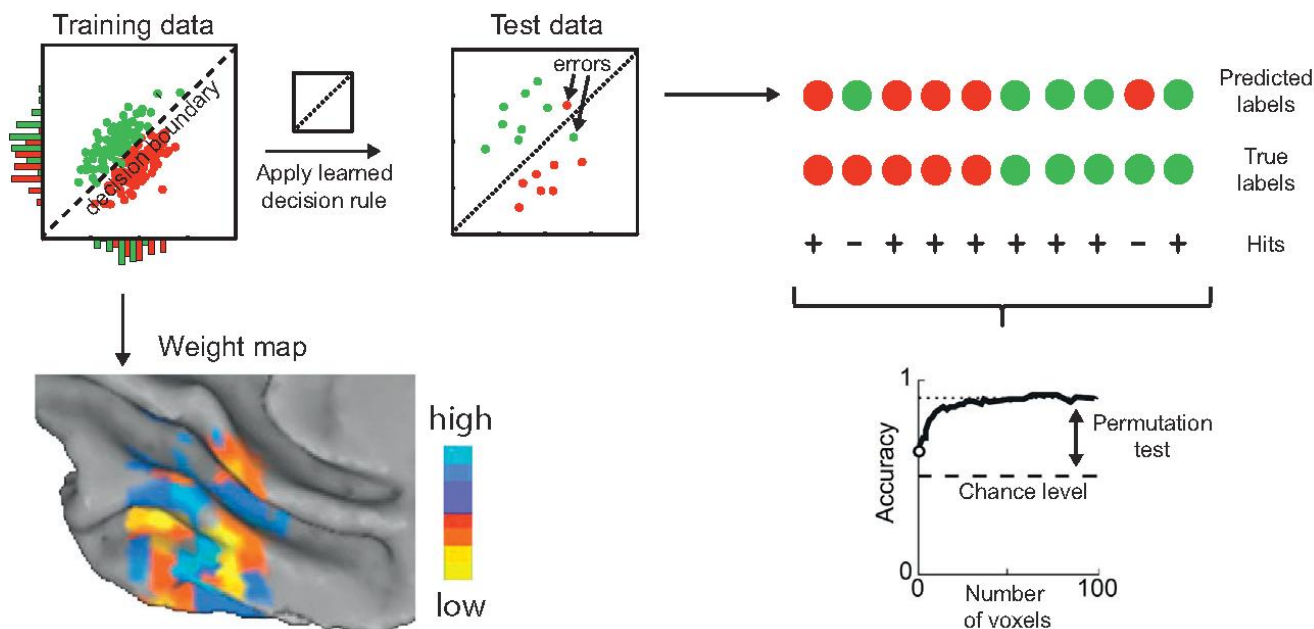
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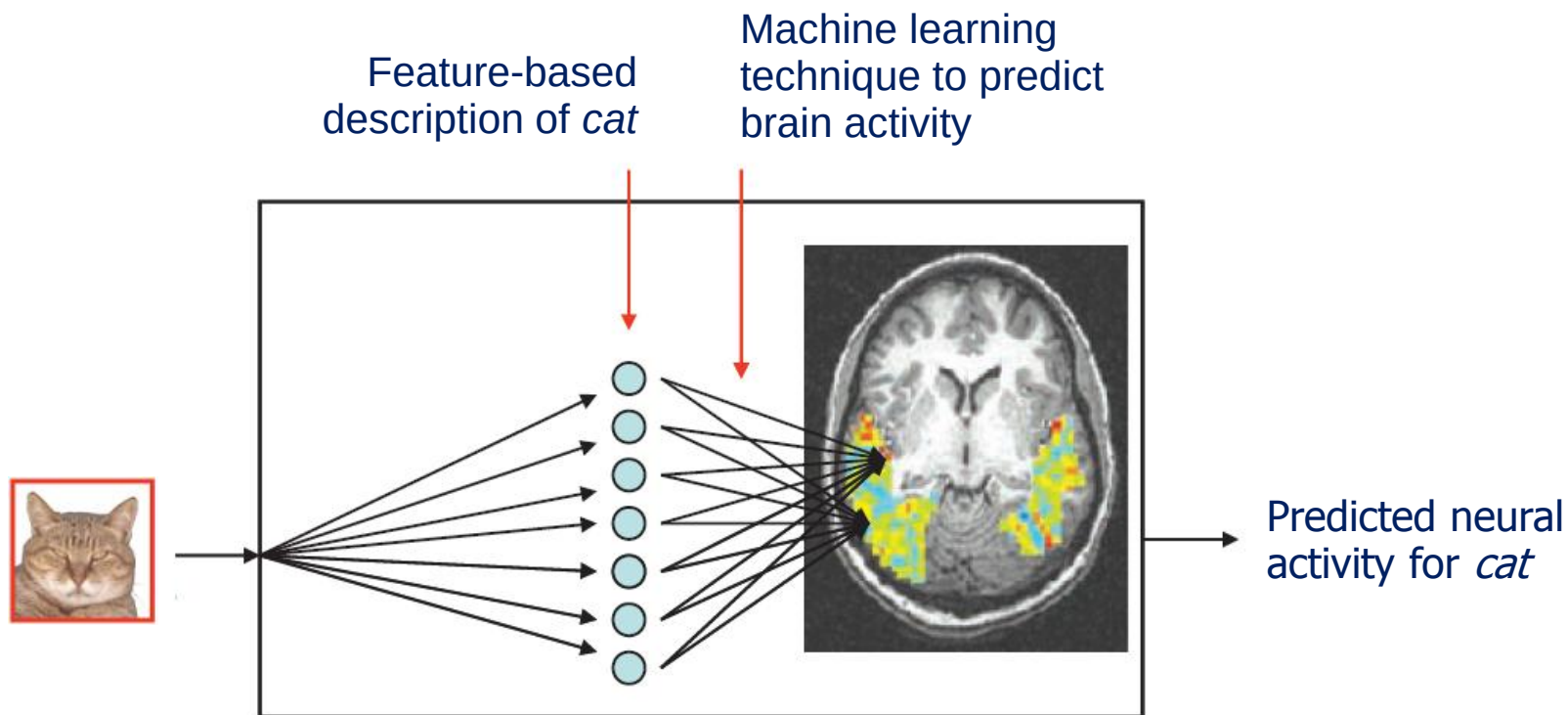
Multivariate specific response to categories



Decoding relies on classifiers and predict stimulus features (e.g., labels) using brain activity



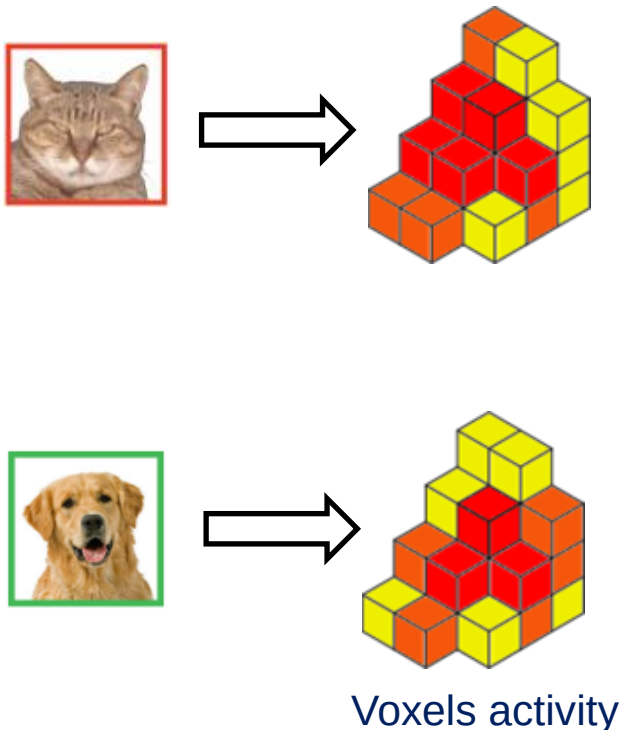
Encoding approaches rely on feature-based models to predict neural activity



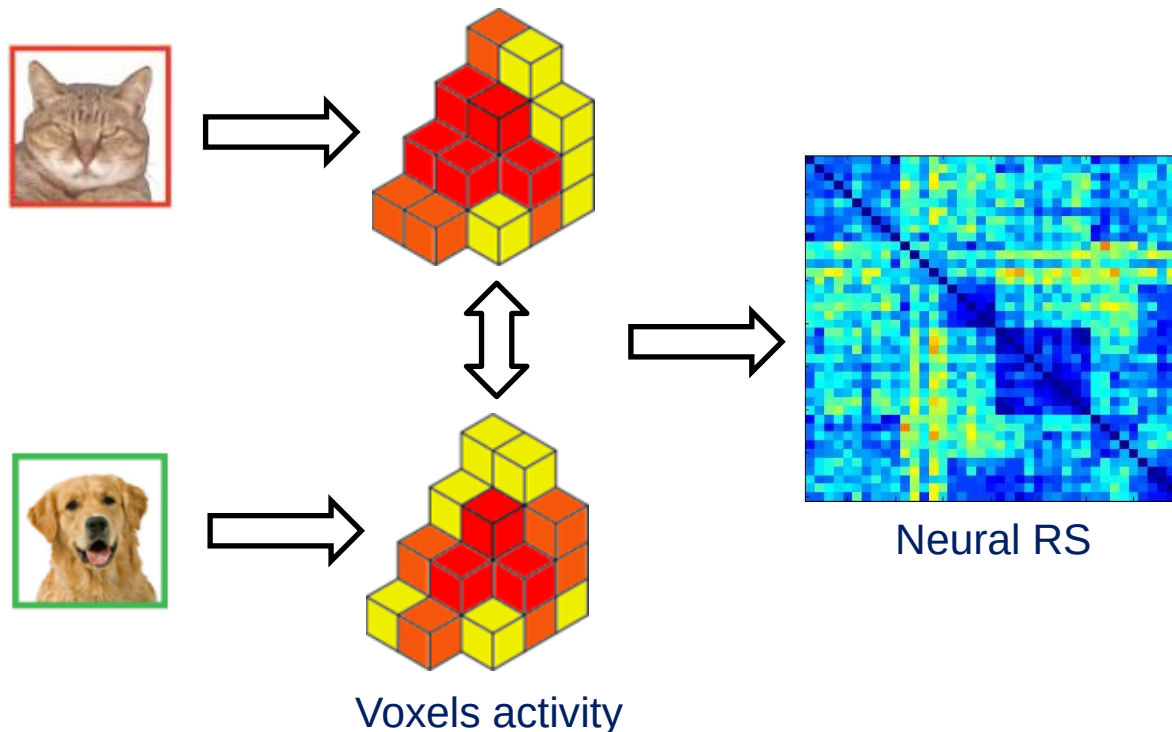
Representational Similarity Analysis (RSA) compares the response patterns of brain activity. RSs are usually adimensional, and easy comparable across models and brain regions



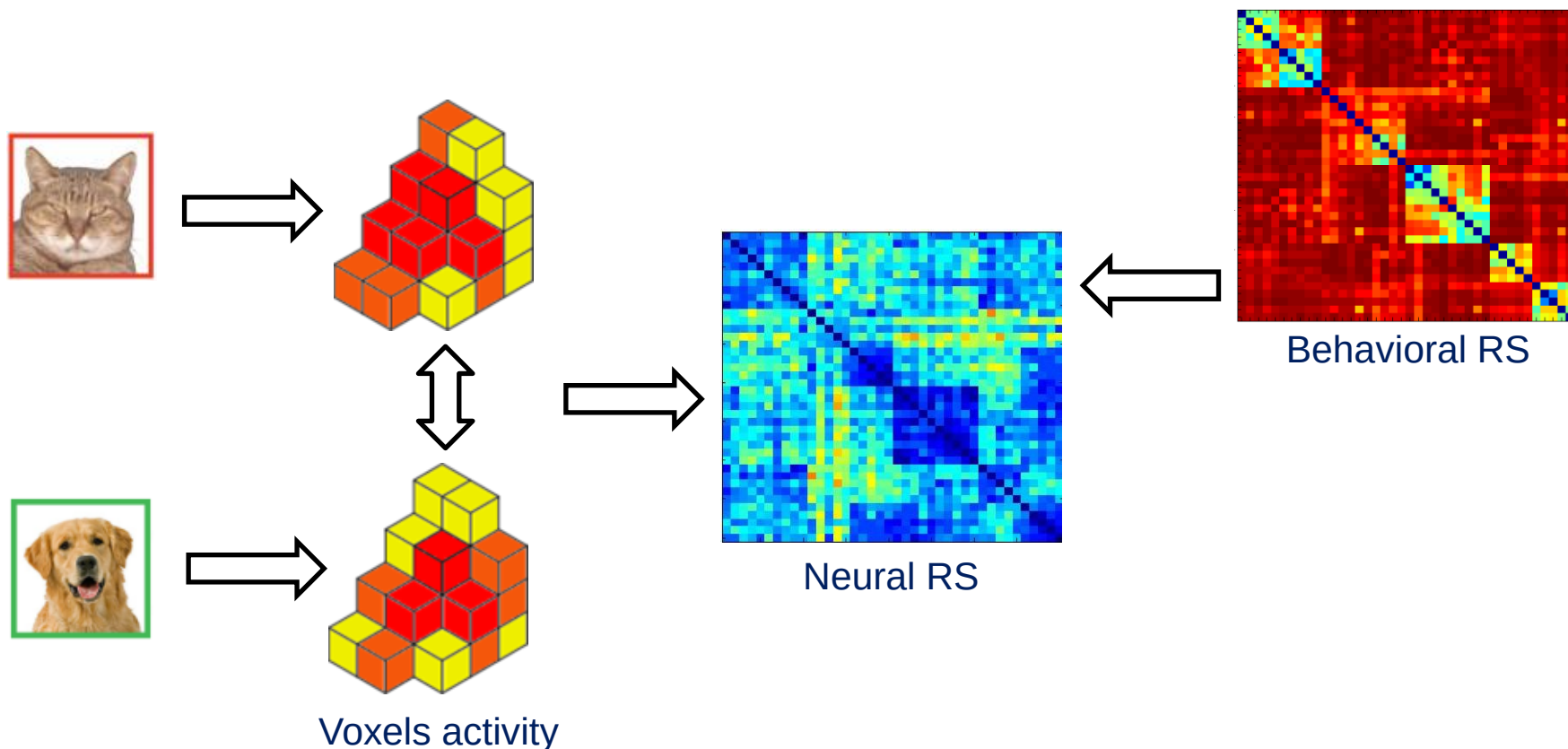
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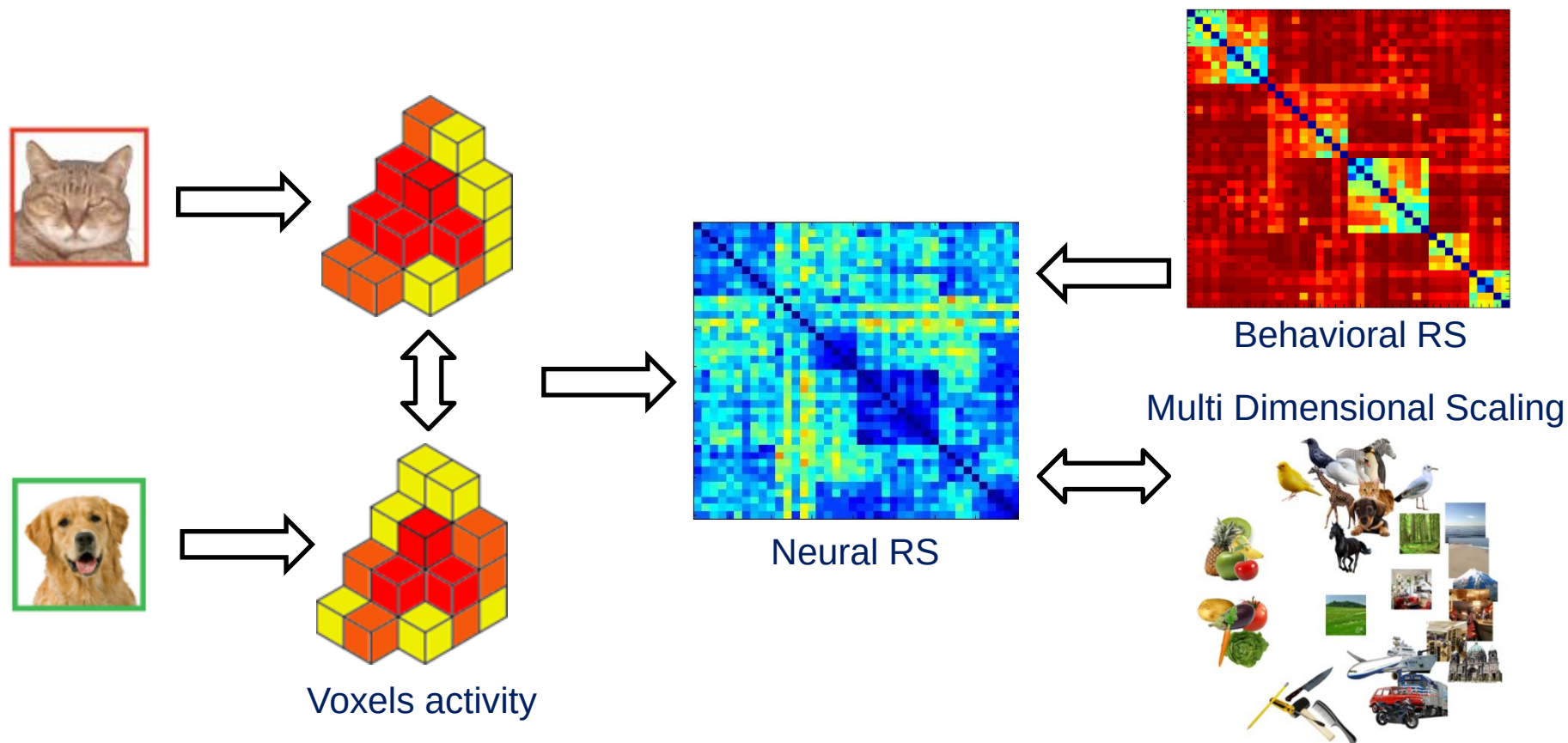
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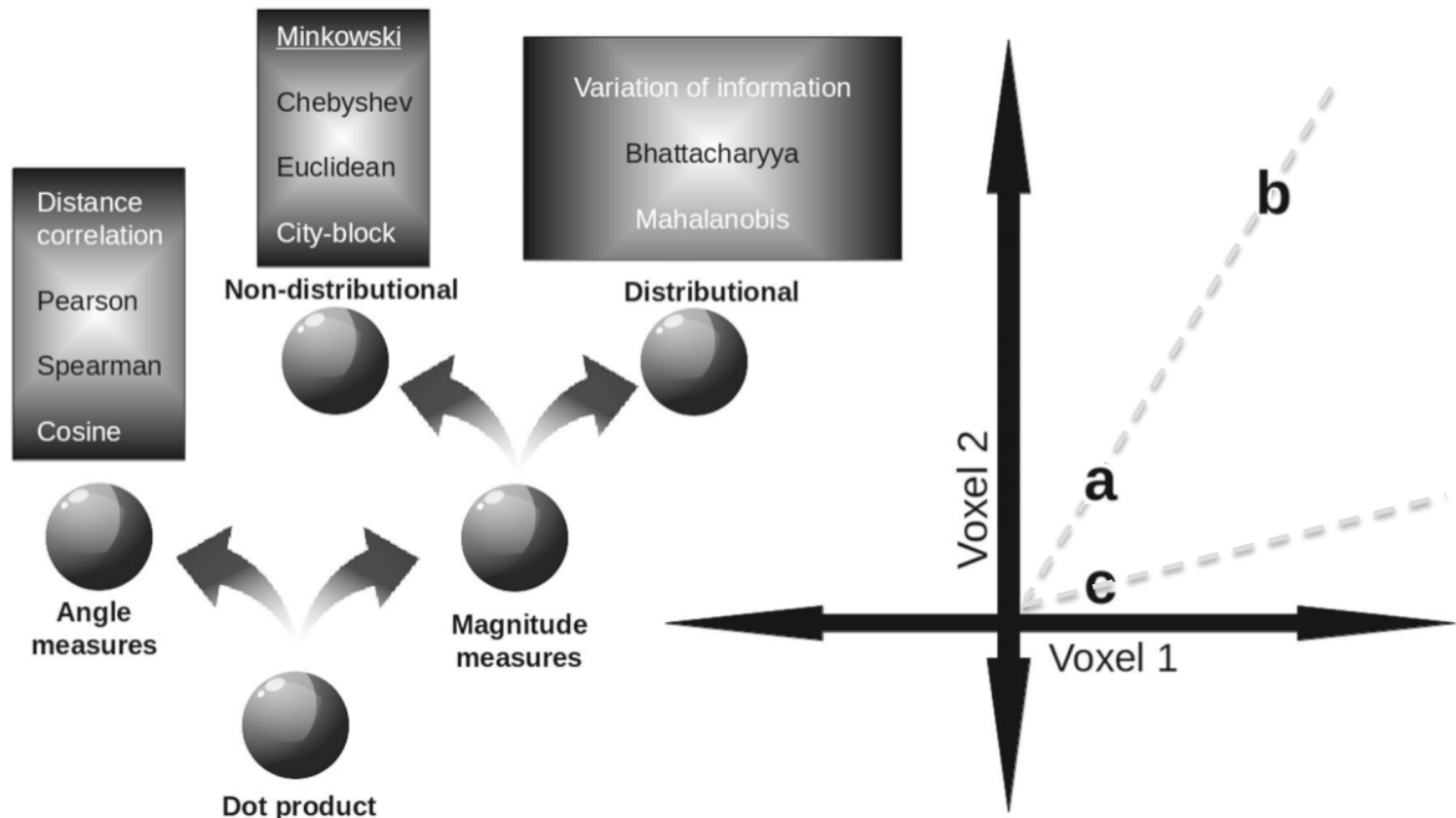
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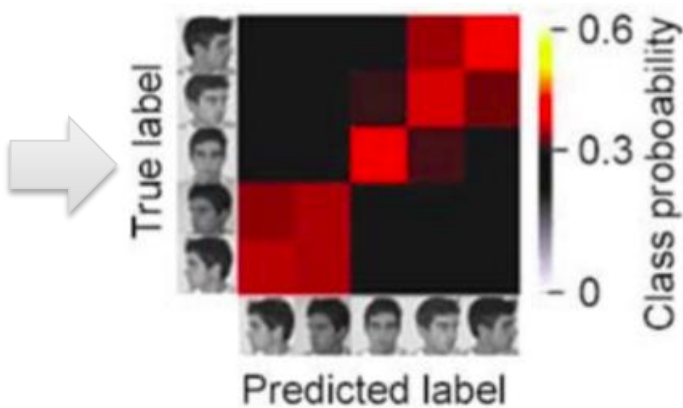
- The choice of the correct distance in a conundrum (Bobadilla-Suarez, 2019; Diedrichsen, 2020)



RSA experimental considerations

- The choice of a metric is not trivial

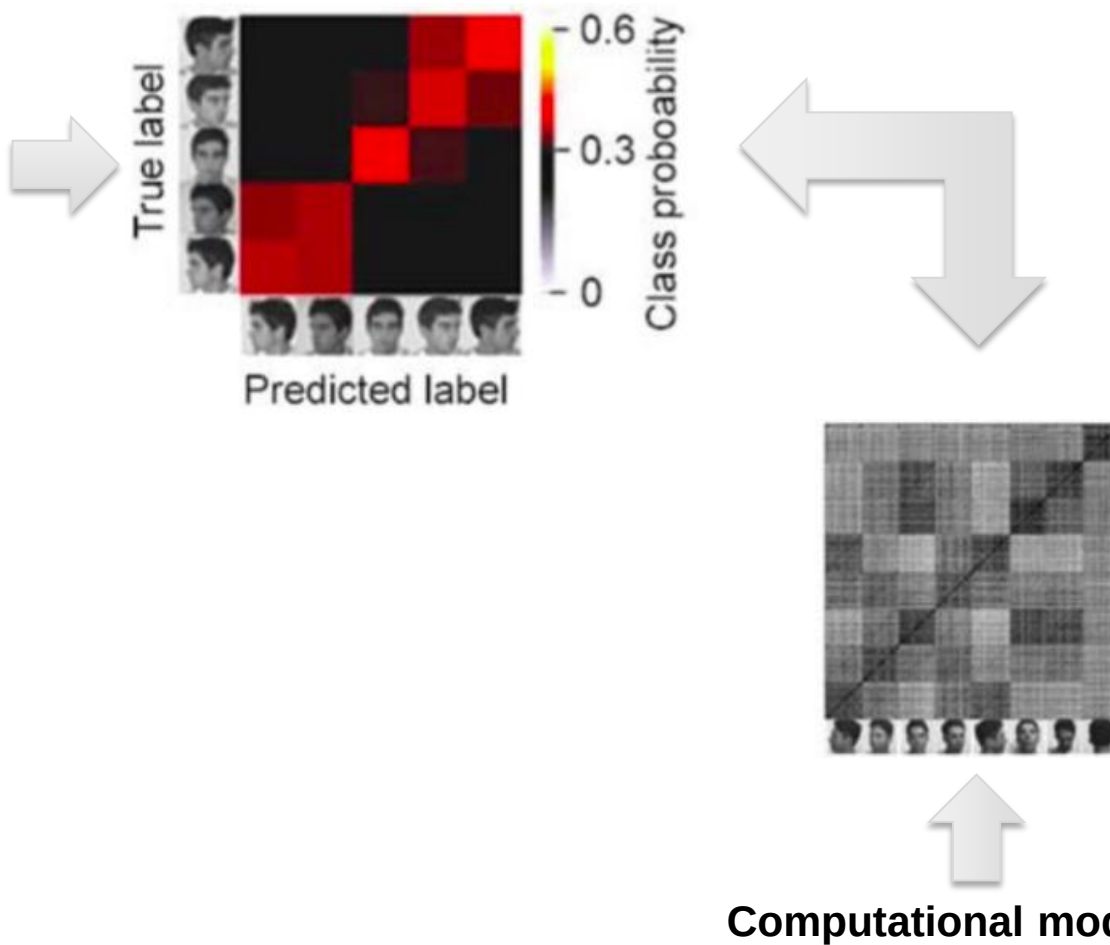
behavioral experiment



RSA experimental considerations

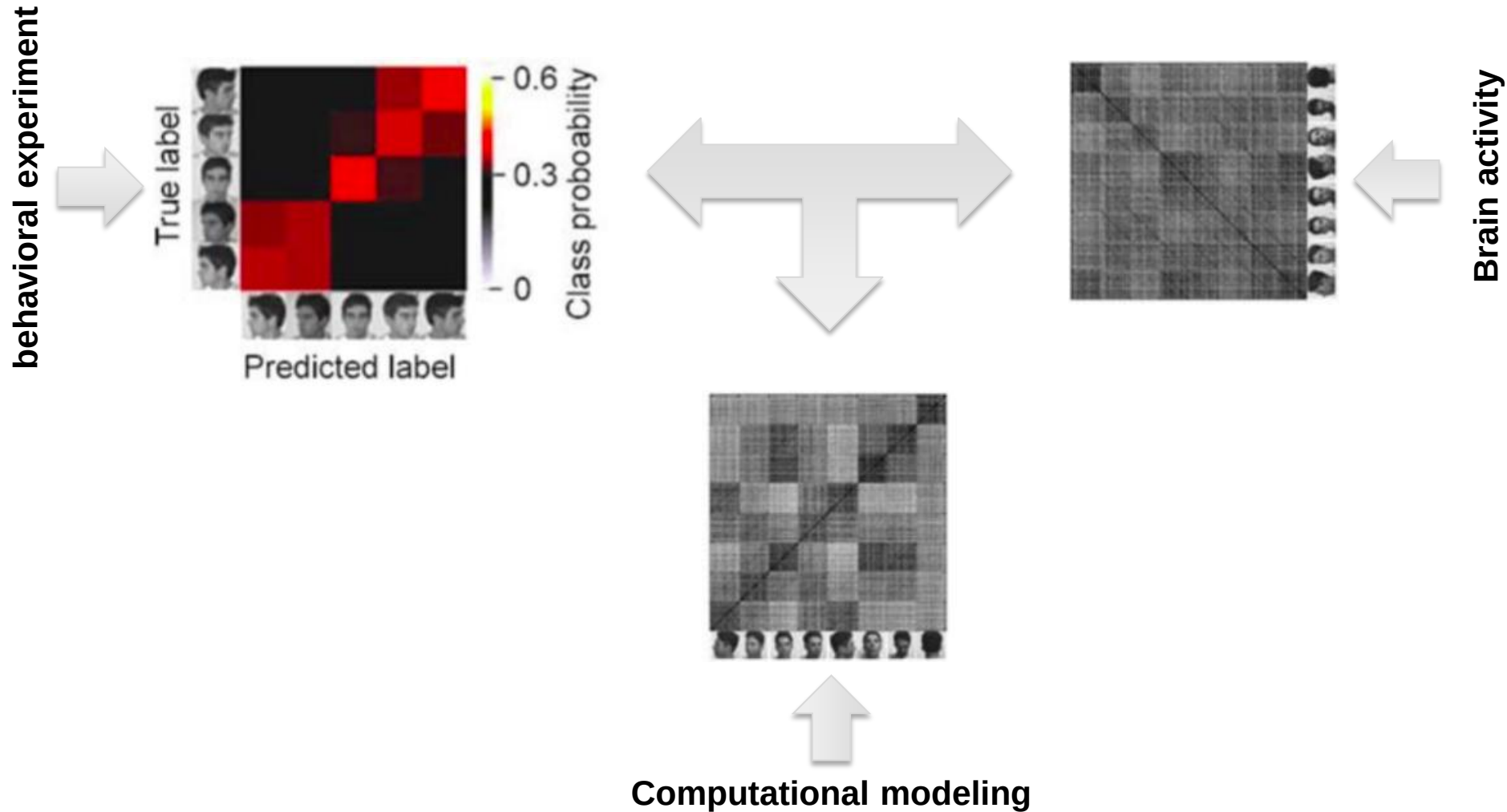
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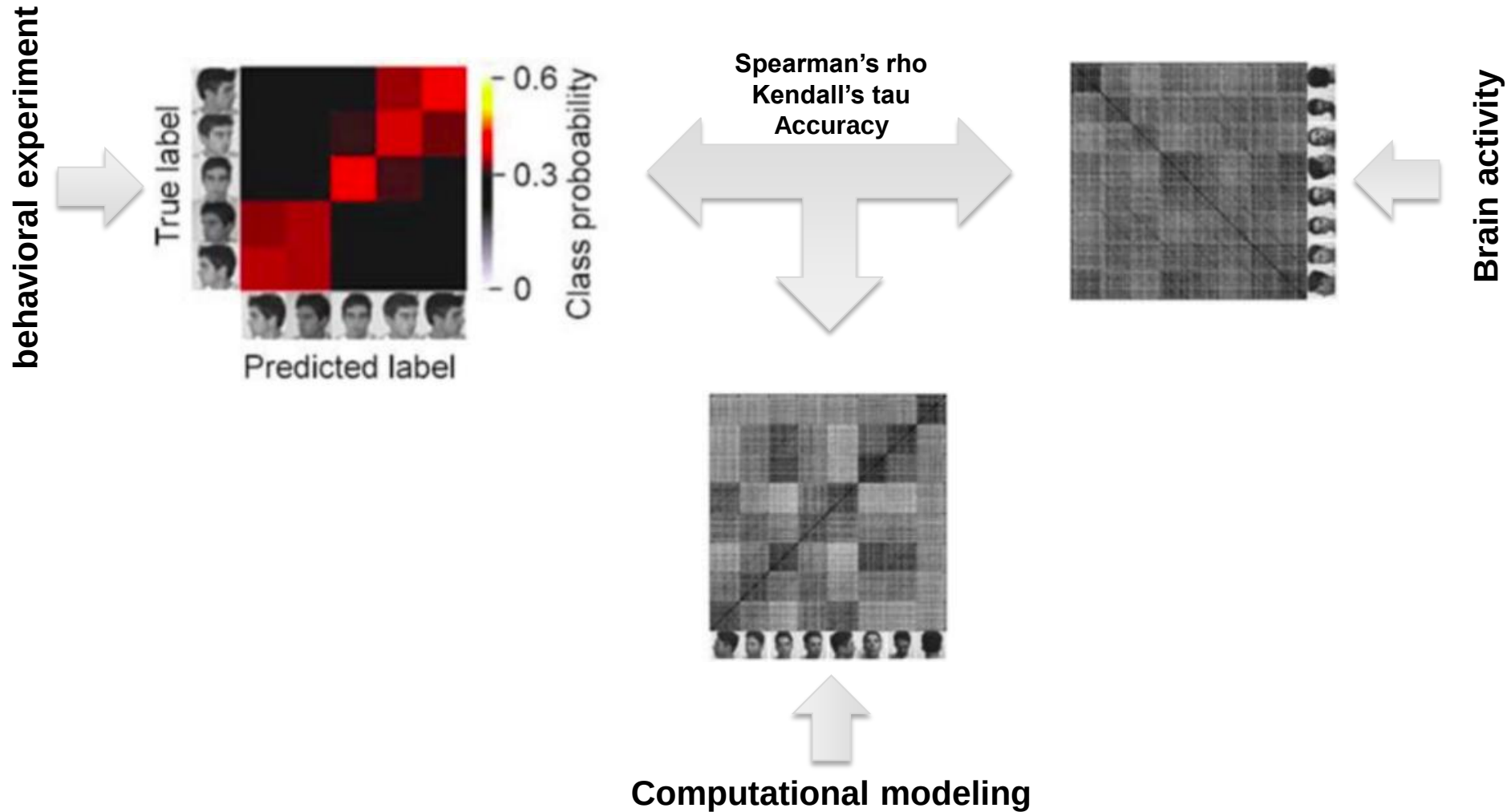
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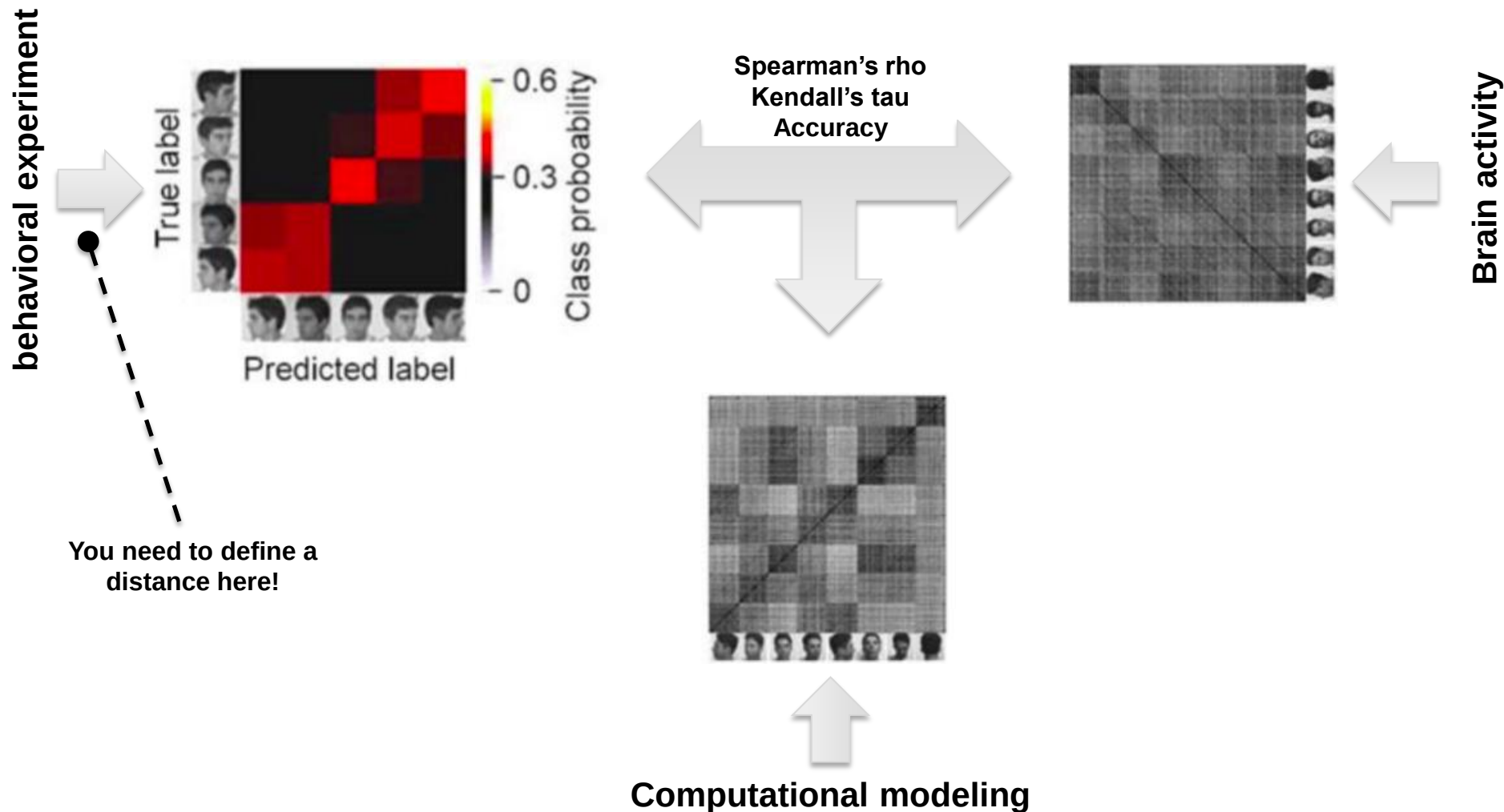
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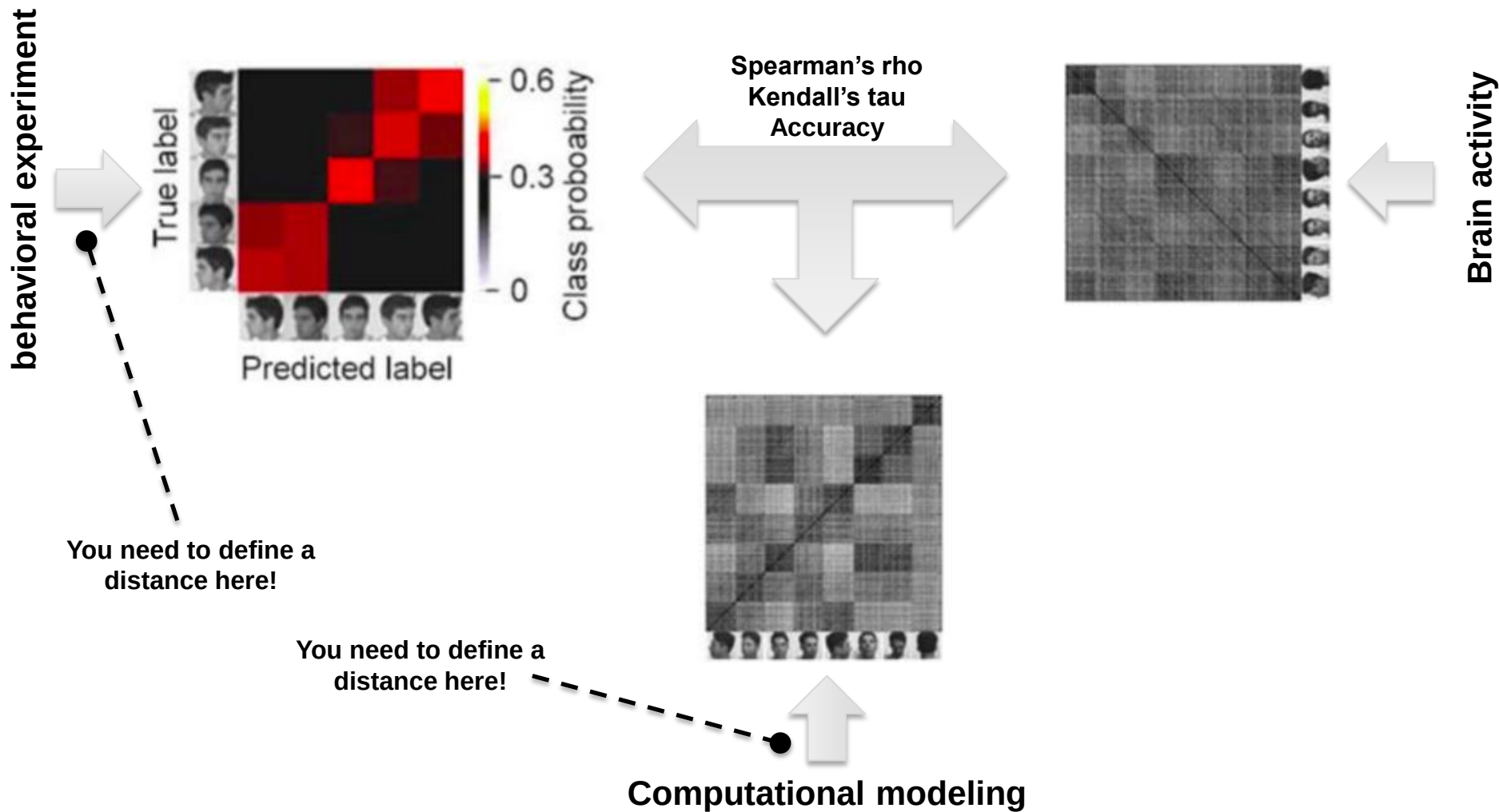
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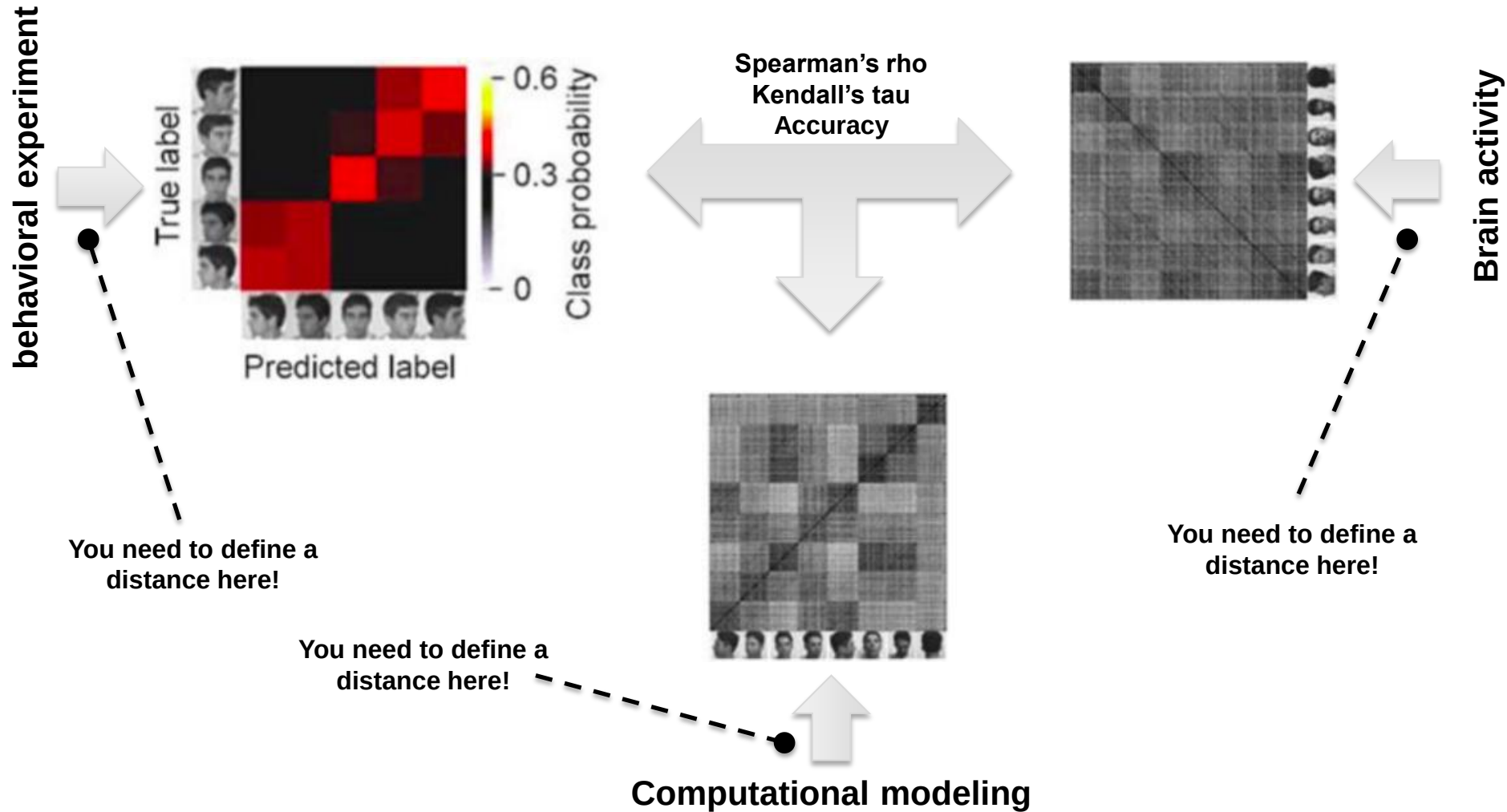
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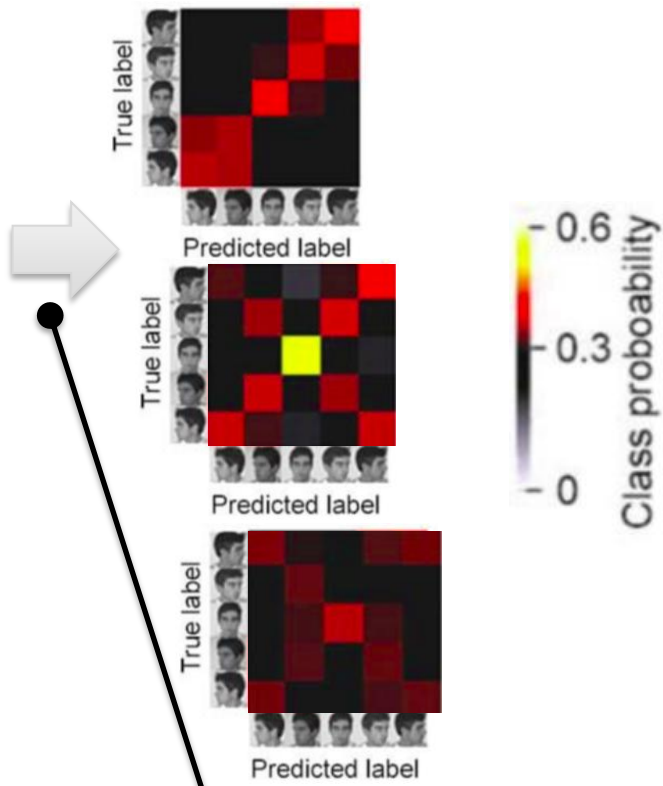
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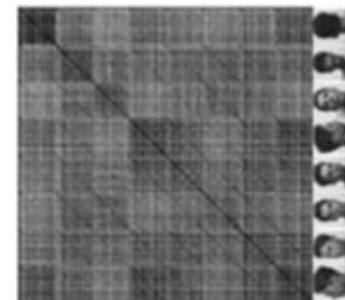
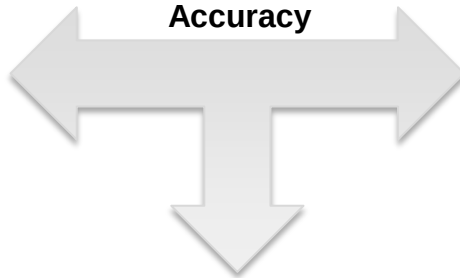
- You are in control of the behavioral experiment!

behavioral experiment

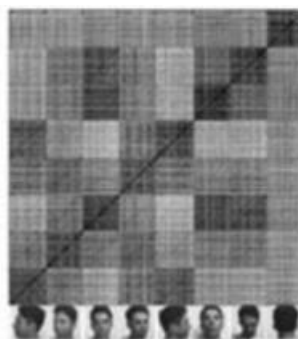


- The similarity matrix should represent the behavioral results
- Task could be tailored to obtain different similarity matrices

Spearman's rho
Kendall's tau
Accuracy



Brain activity

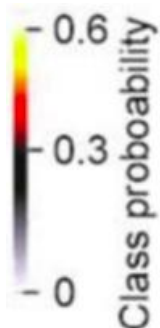
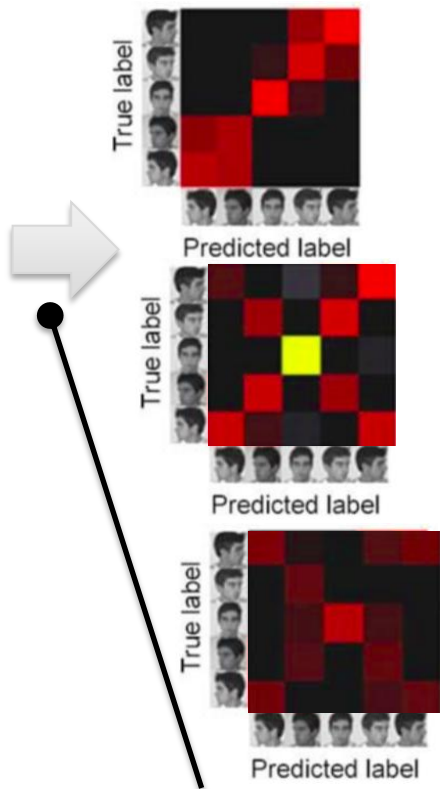


Computational modeling

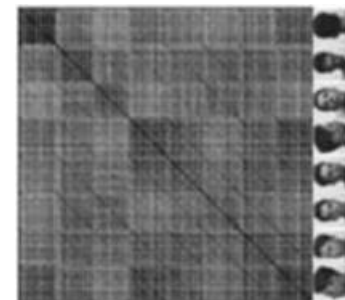
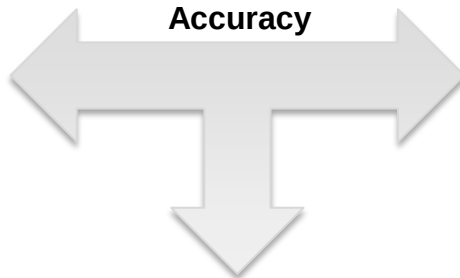
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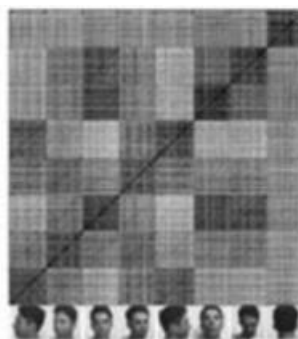
behavioral experiment



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Brain activity

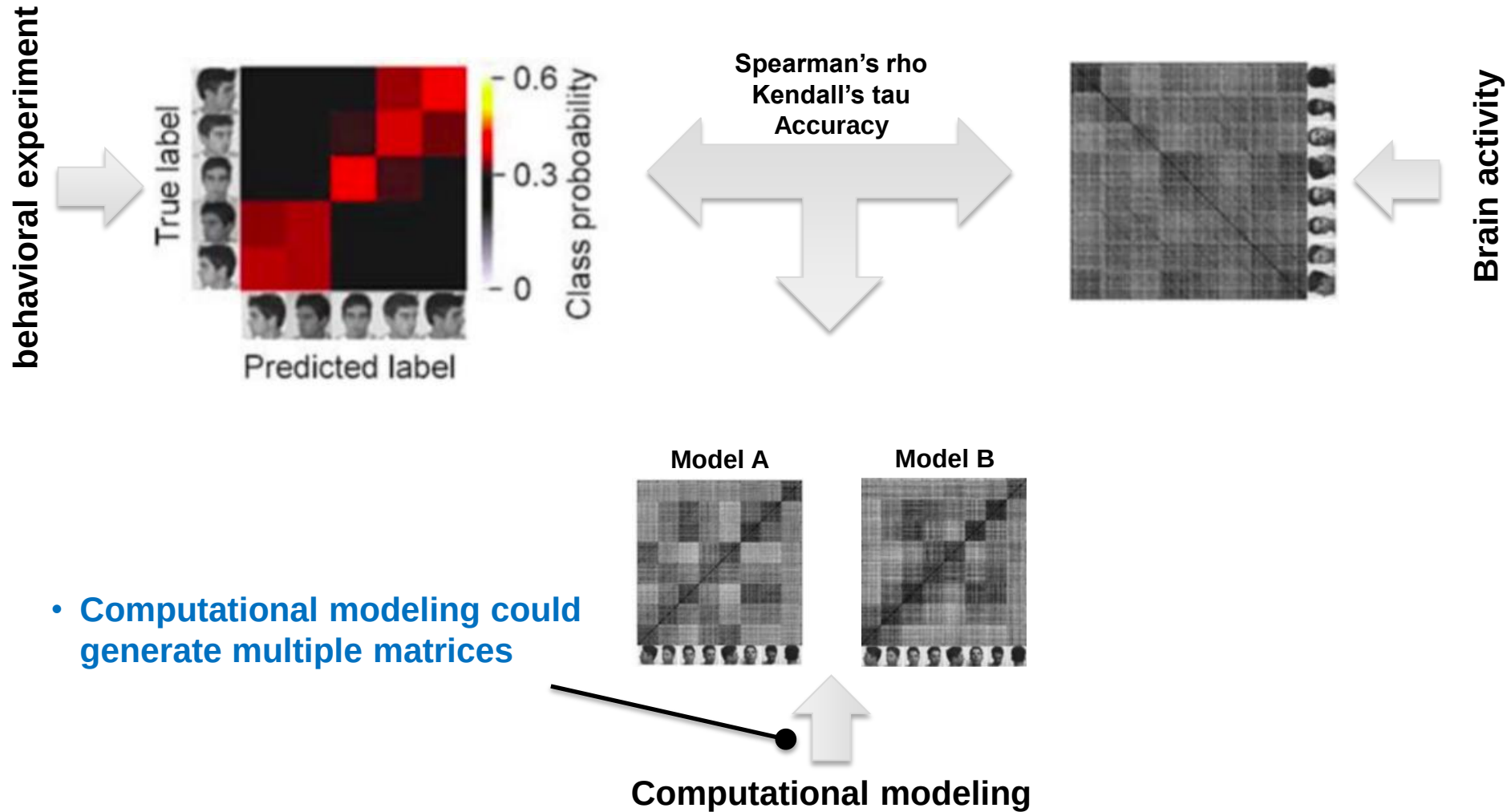


Computational modeling

- The behavioral matrix could be even a confusion matrix or a similarity-based one

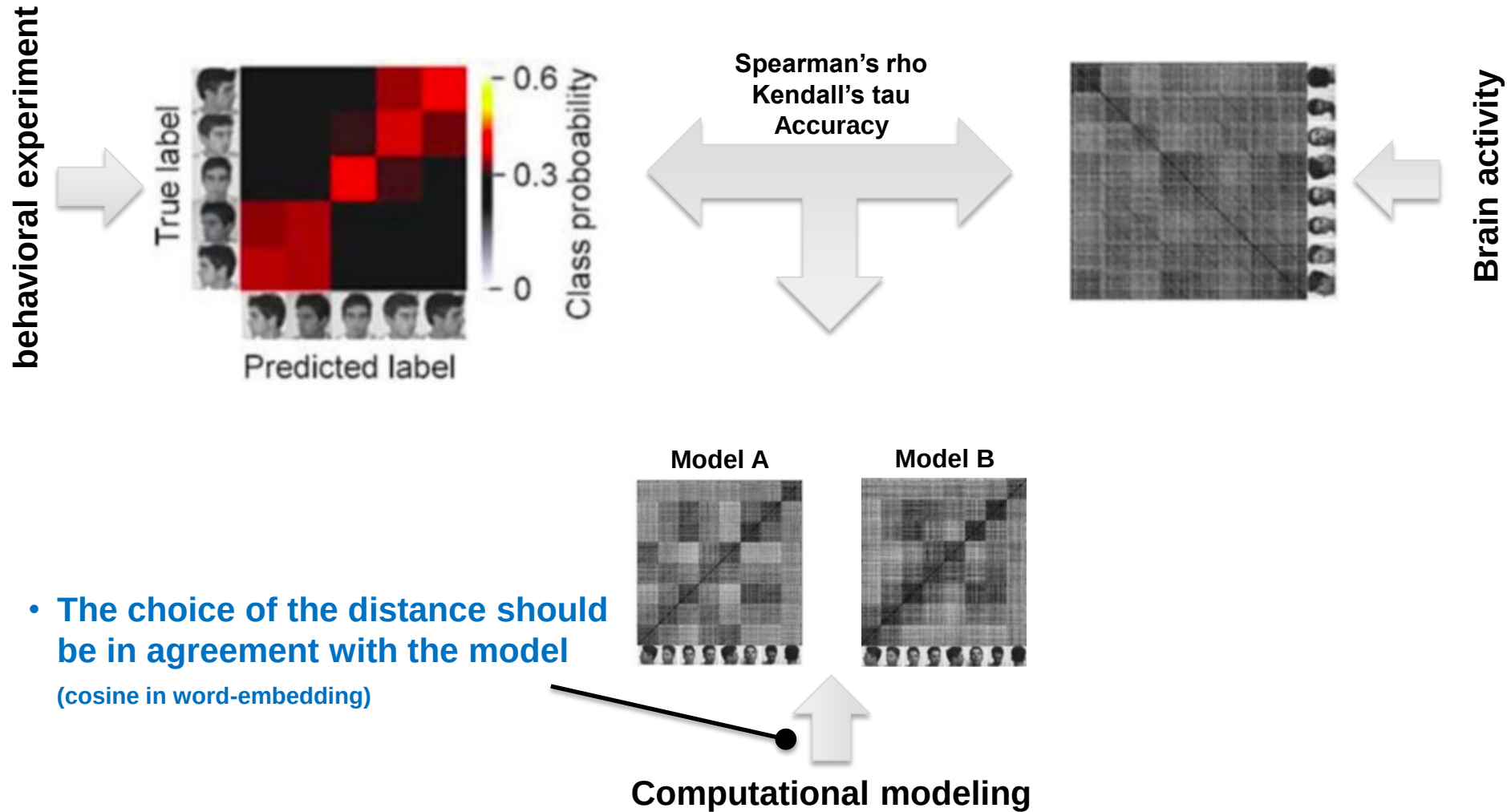
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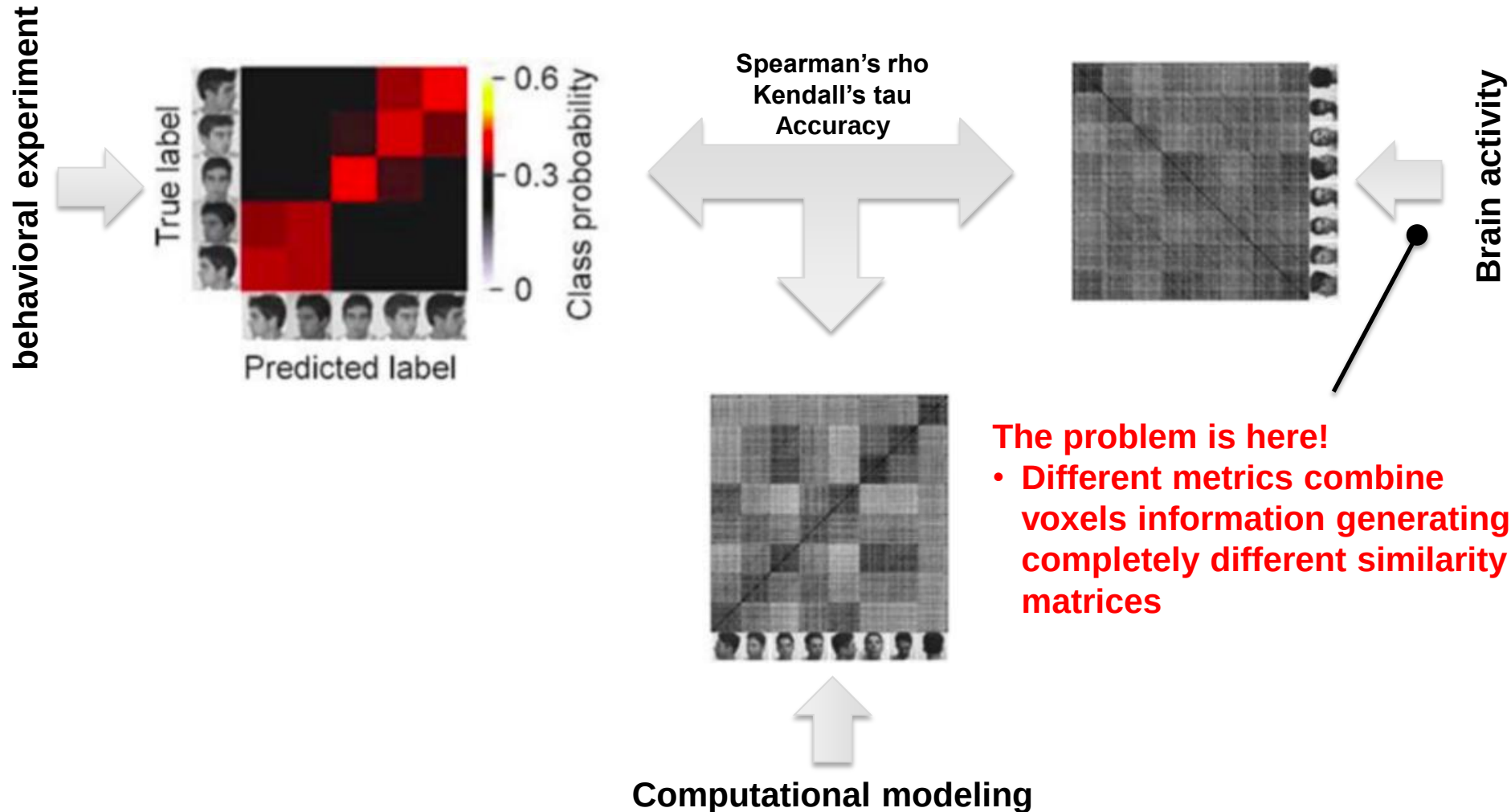
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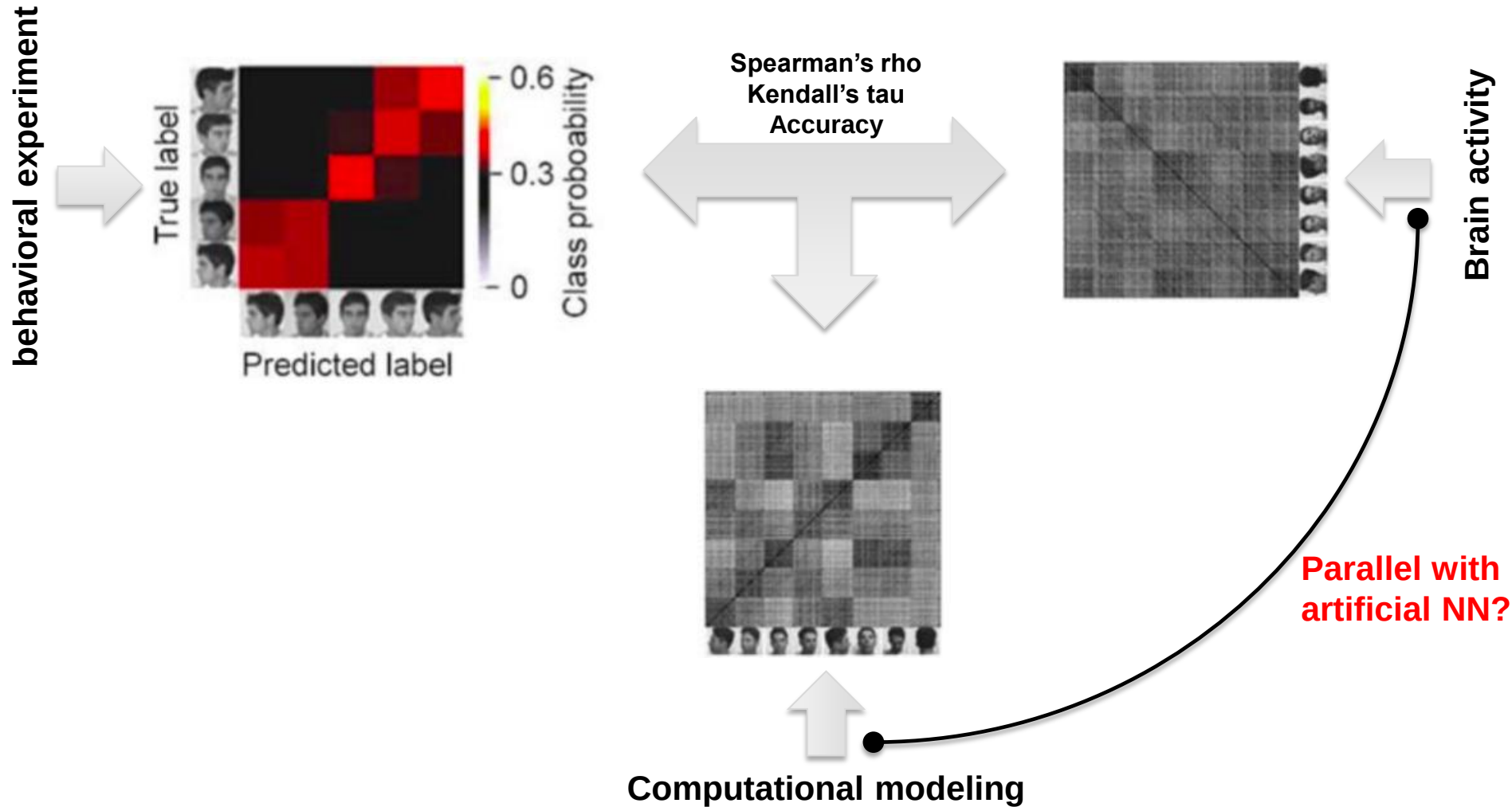
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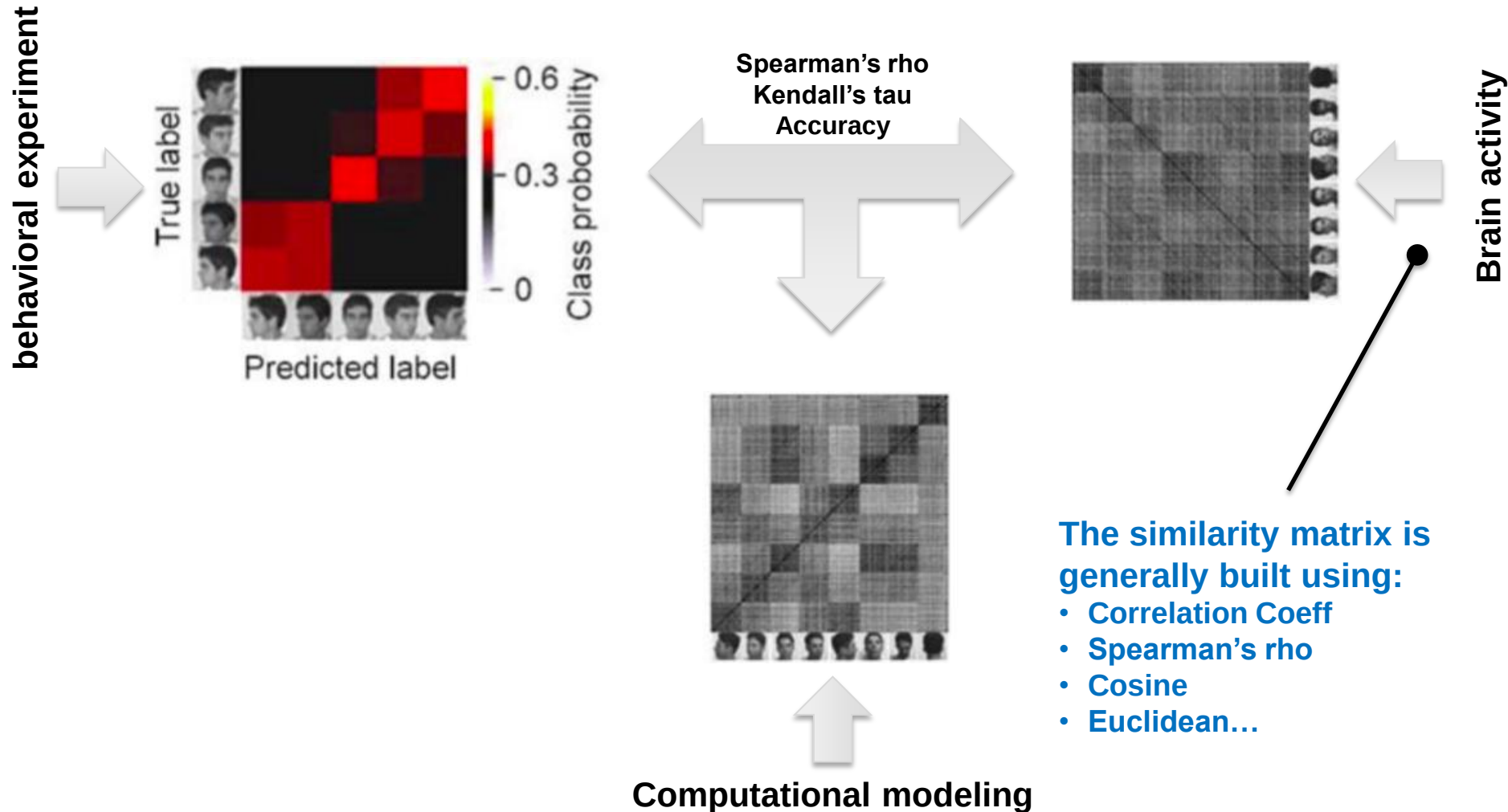
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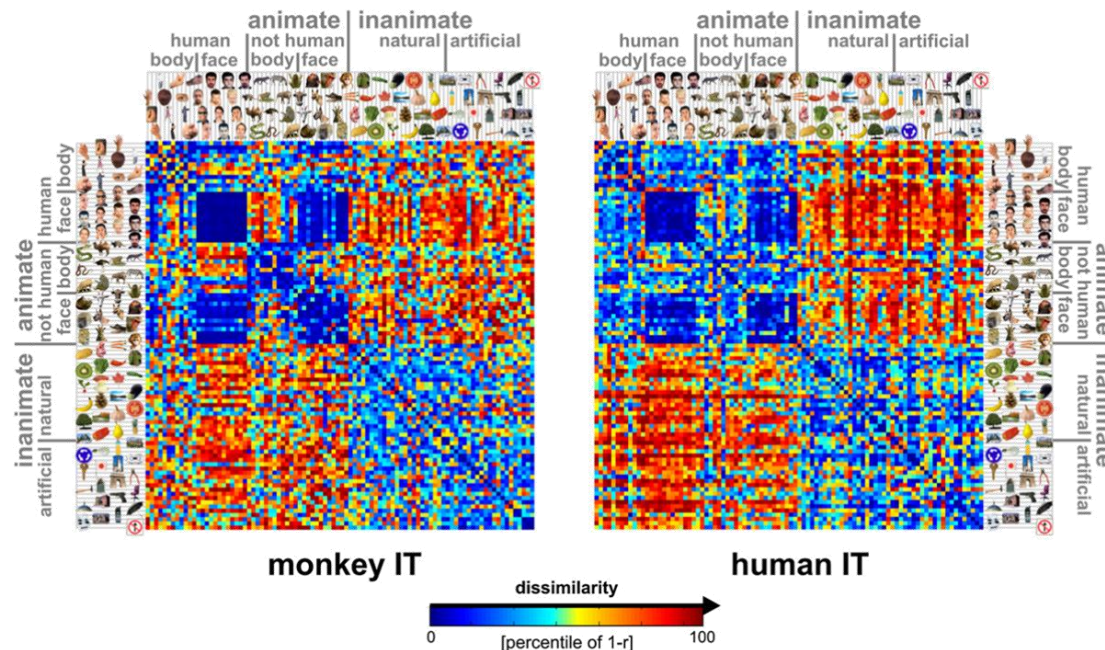
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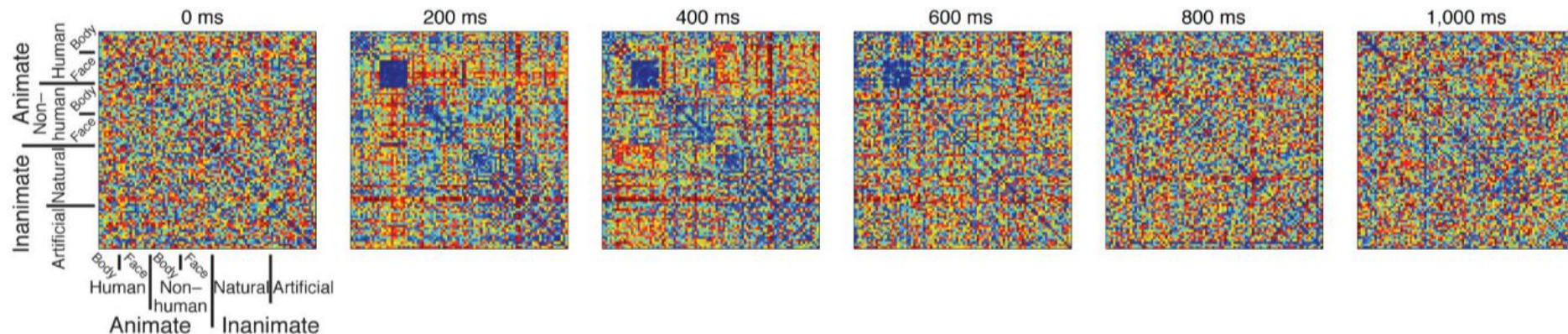


- Neural signal extracted from a region of interest, generally a value for each voxel and stimulus to generate a RDM from brain activity
- Stimuli must be characterized according to a-priori descriptions (e.g., class membership) or using computational models (e.g., GIST for vision, power-spectrum for sounds, linguistic features for concepts) to generate a RDM from the experimental materials
- Stimuli must be provided in a random order with an adequate design (Mumford, 2014)
- (critical) Methodological choices are the procedures to build the two RDMs (i.e., distance measures) and how to select the ROIs

- RSA could merge and compare representations from different sources (Kriegeskorte 2008; Kriegeskorte 2008b; Kriegeskorte 2019)
- RSA generally build a RDM in each subject avoiding direct matching of voxels: this is advantageous for smaller ROIs or with a small voxel size (<2mm)

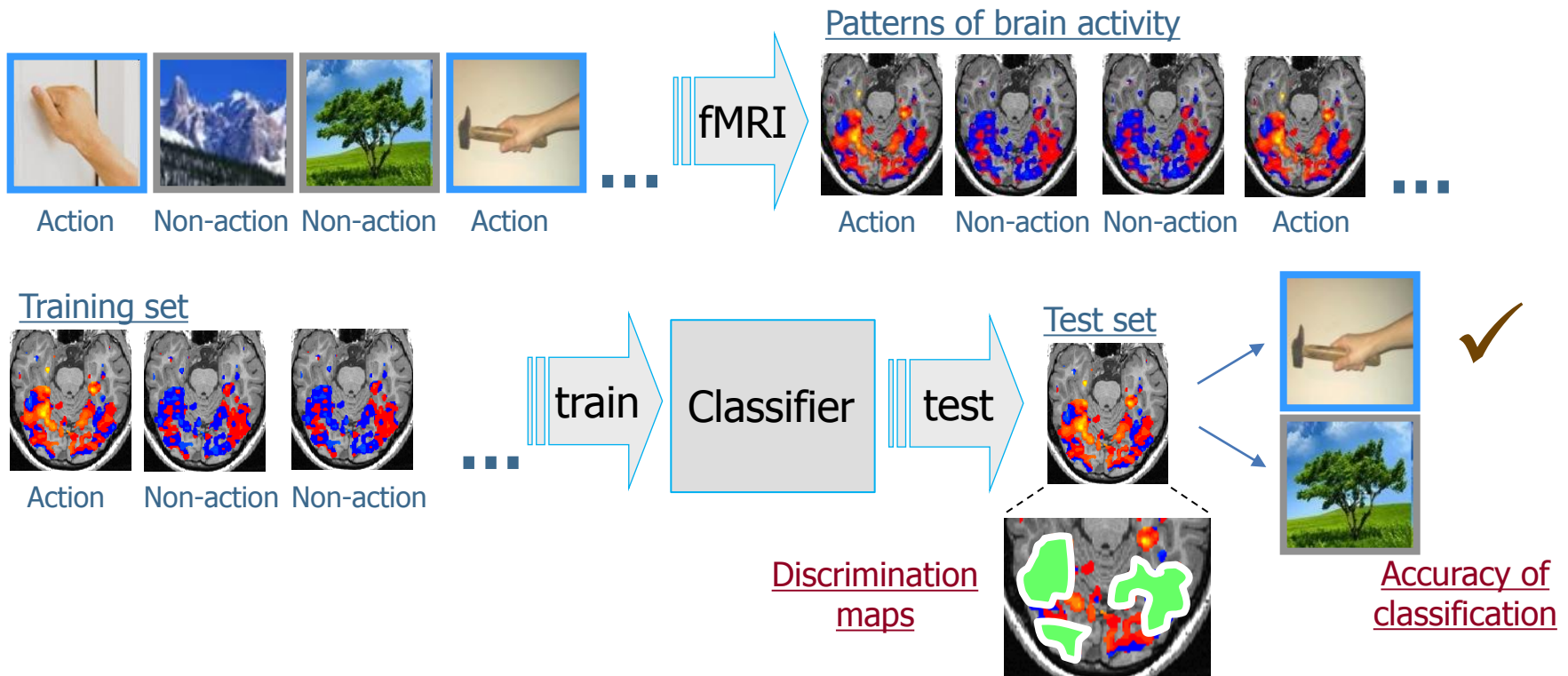


- As always, the selection of the ROI is crucial!
- If you have the *time dimension* (MEG, EEG), you may build RDMs in time (Cichy, 2014)



- RSA allows to test multiple models with different dimensionality/complexity (Khaligh-Razavi, 2014)

Decoding approach



Decoding consists of few simple steps:

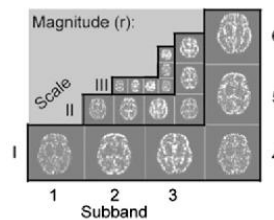
1. Have a clear a-priori subdivision of the stimulus set in few classes (2 or more)
2. Multiple repetitions of each stimulus category
3. A ROI
4. Split the dataset in training and test sets (with small datasets, consider to nest train&test within a cross-validation loop)
5. Train a classifier (e.g., LDA, SVM, ANN, k-NN) with the training set, and test its performance with the test set
6. Obtain accuracy **and** alternative measures (e.g., sensibility and specificity for the binary classifier)
7. Perform a permutation test (e.g., label shuffling) to generate a null-distribution of accuracies

Problems: the number of voxels

- Decoding algorithms operate on a matrix of stimuli by voxels. It is mandatory to have a higher number of stimuli as compared to the voxels (stimuli >> voxels) to avoid over-fitting of noise (a.k.a., curse of dimensionality)
- Few algorithms (e.g., SVM) are more resilient to the curse of dimensionality
- Common strategies (to survive) include:
 - Whole-brain approaches: use Recursive Feature Elimination (RFE) to prune irrelevant voxels (still require lots of stimuli and you need to split the data in three sets, train, test and validation) or more aggressive solution (e.g., dimensionality reduction such as PCA, ICA)
 - Parcellate the cortex in large-to-small ROIs or in small ROIs directly
 - Use a searchlight approach



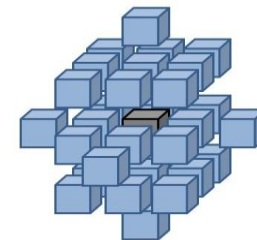
whole-brain



Large-to-small



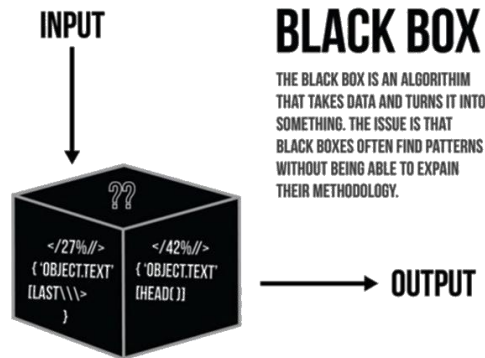
Atlas

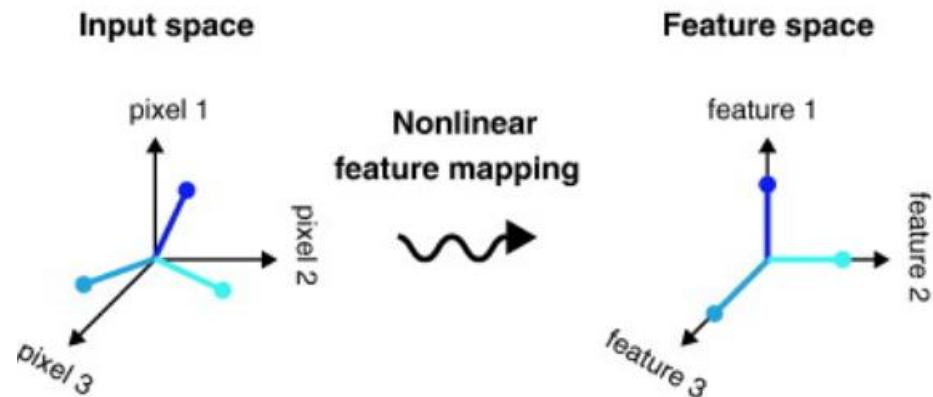


Searchlight

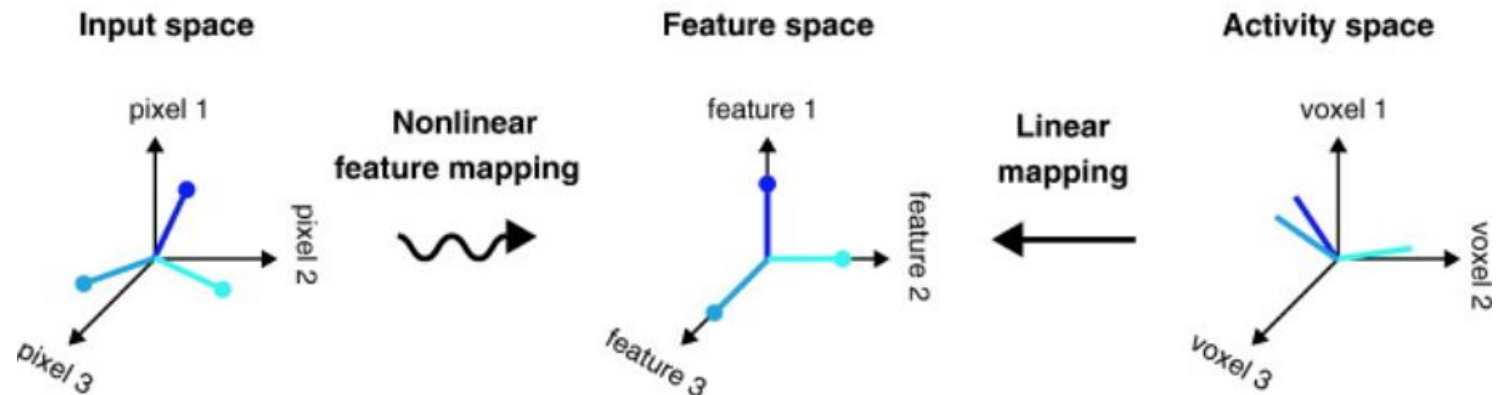
Problems: the interpretation!

- Classifiers are black-boxes! By pruning voxels or mapping weights, it is possible to highlight the most important patches of cortex involved in the classification, but the underlying neural code (i.e., algorithms and encoded features) remains obscure!
- Decoding remains of utmost interest for the identification of clinical biomarkers (e.g., binary classifiers of patients vs controls)
- Decoding is still a valid option when the researcher has no clue on the features characterizing the experimental classes, so that RSA or encoding procedures could not be used



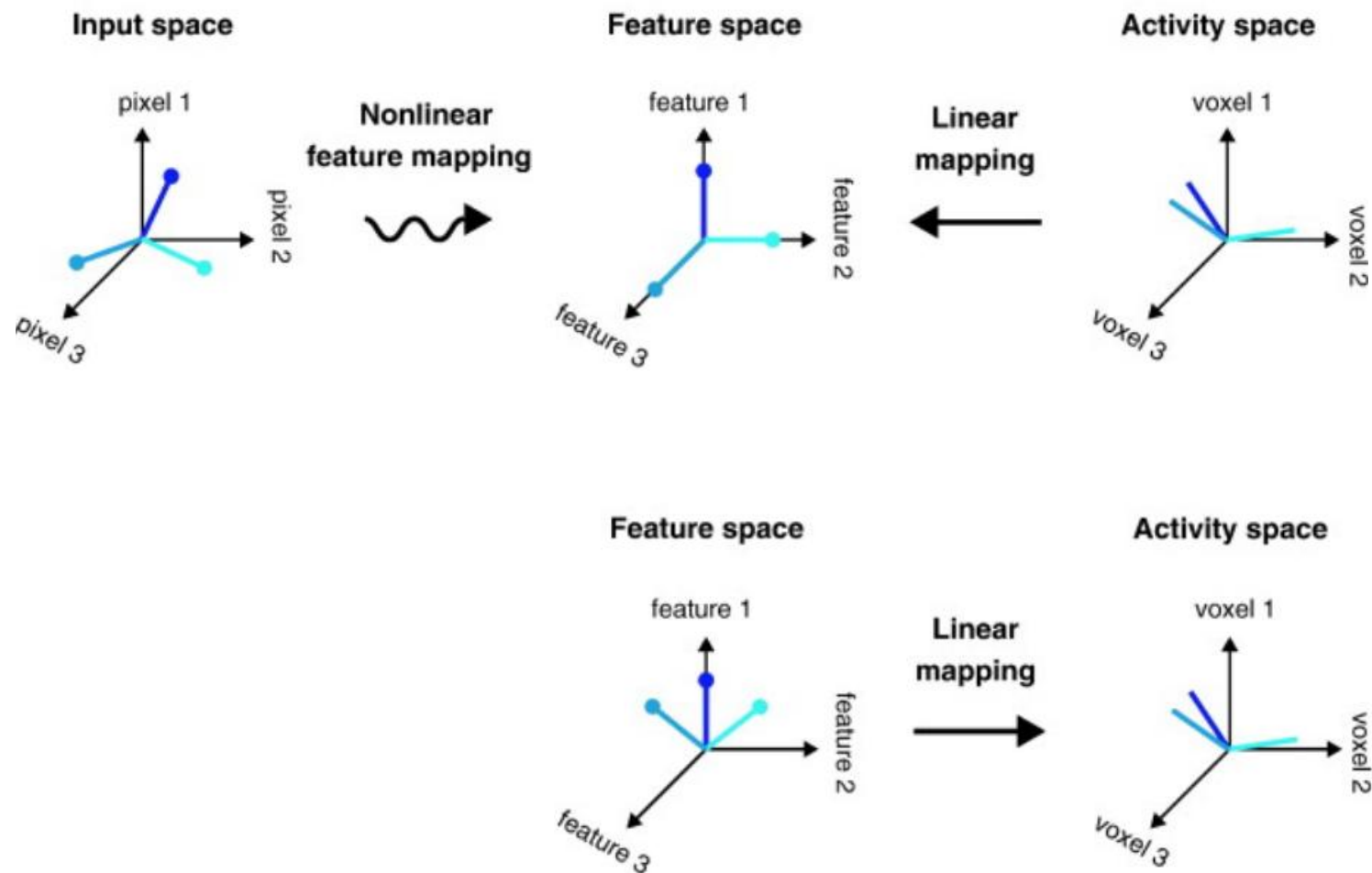


Encoding approach



DECODING

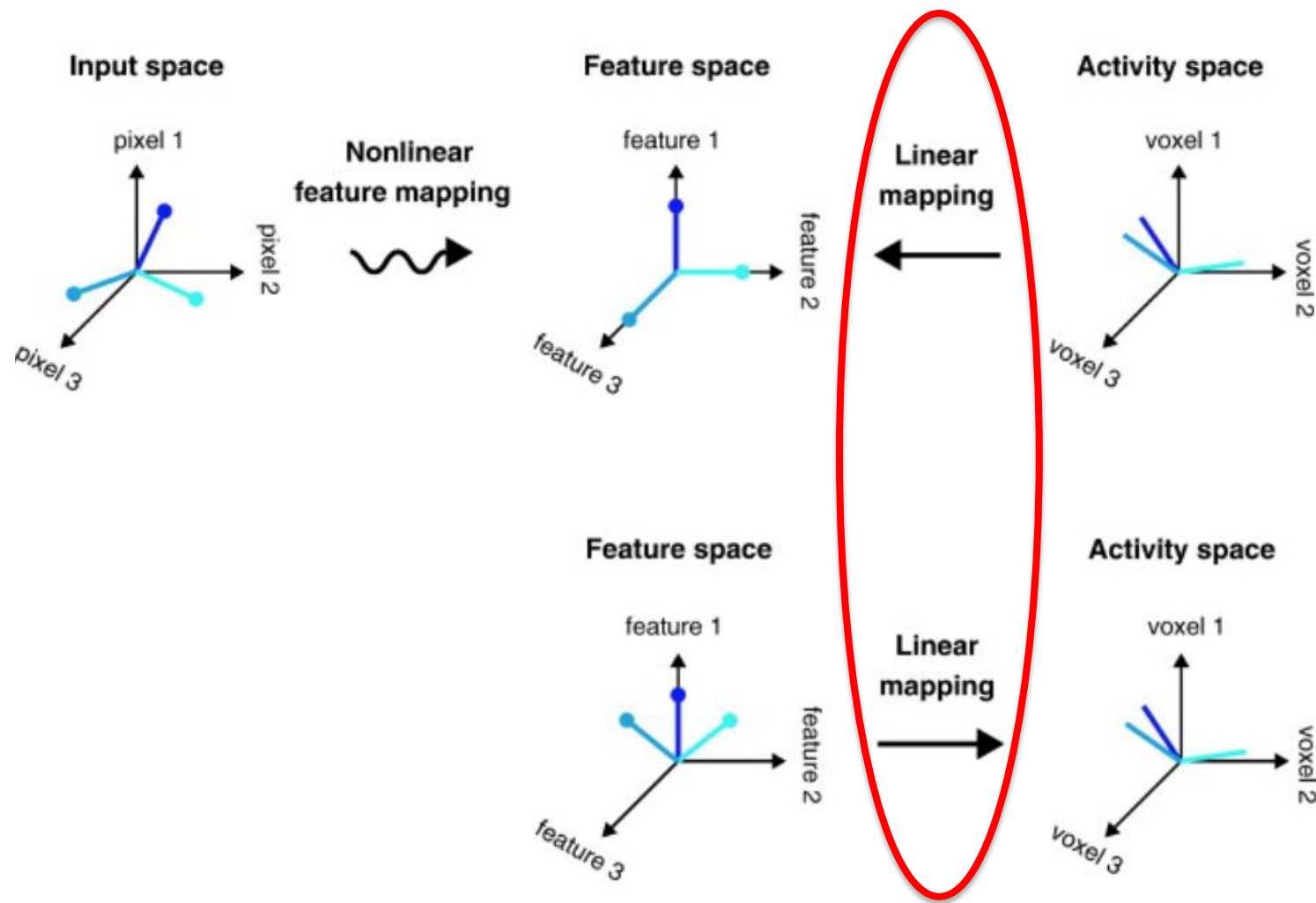
Encoding approach



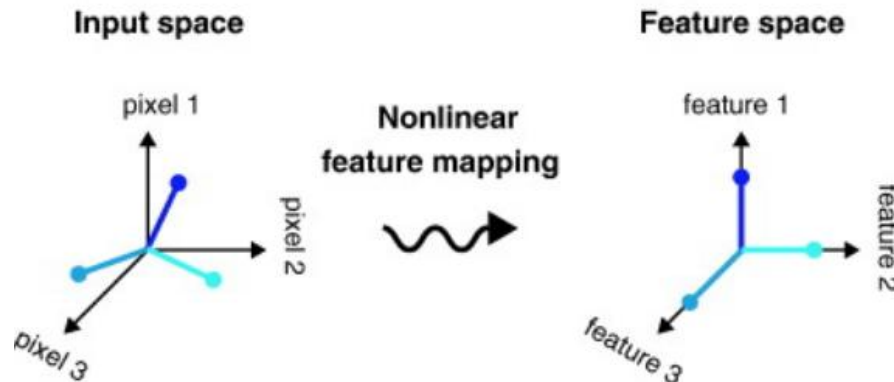
DECODING

ENCODING

Encoding approach



- To perform encoding you need to describe your stimuli by means of a computational models to obtain a matrix of stimuli*features
- Using naturalist continuous stimulation, computational models generate a matrix time*features
- Pay attention to the number of features extracted: similarly to decoding, the number of stimuli (or timepoints) should be $\ll$ of the number of features

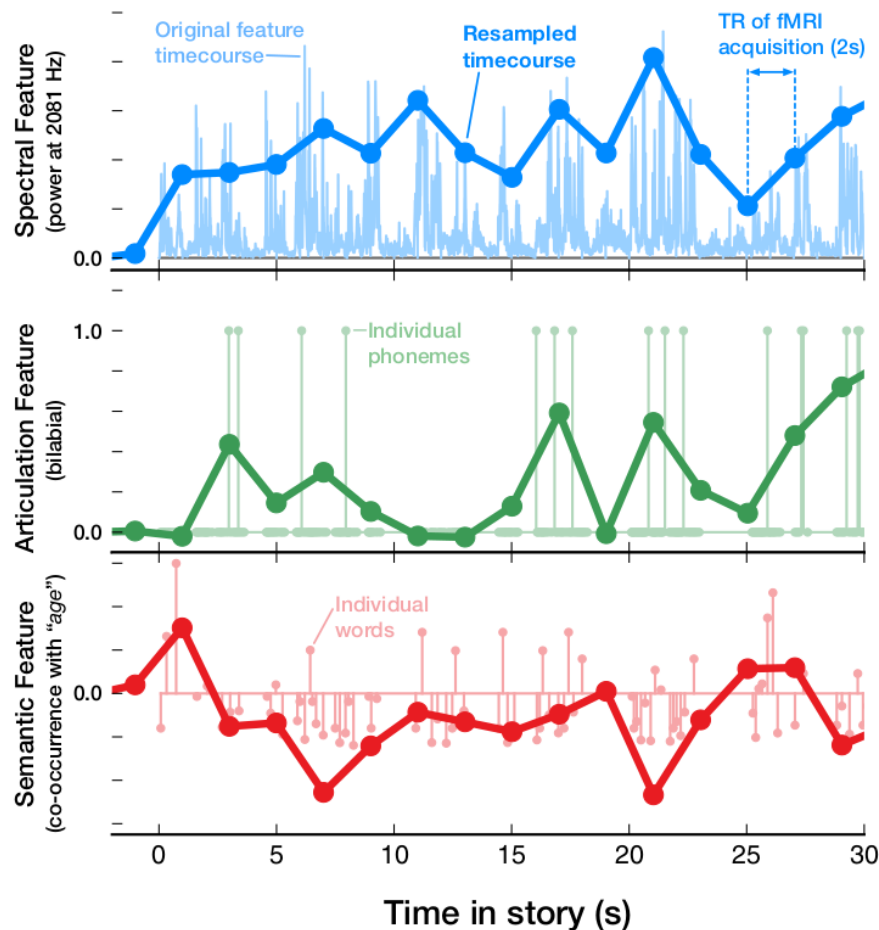
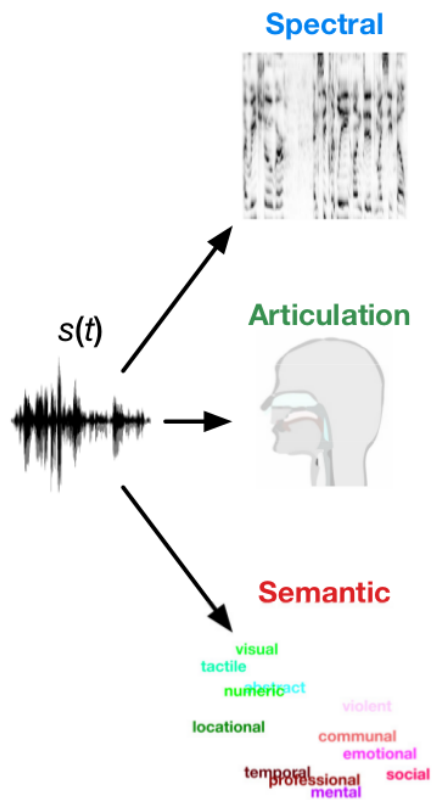


Sound
Waveform

Feature
Spaces

Downsampled Feature
Representations

Feature
Matrices



Frequencies (80)

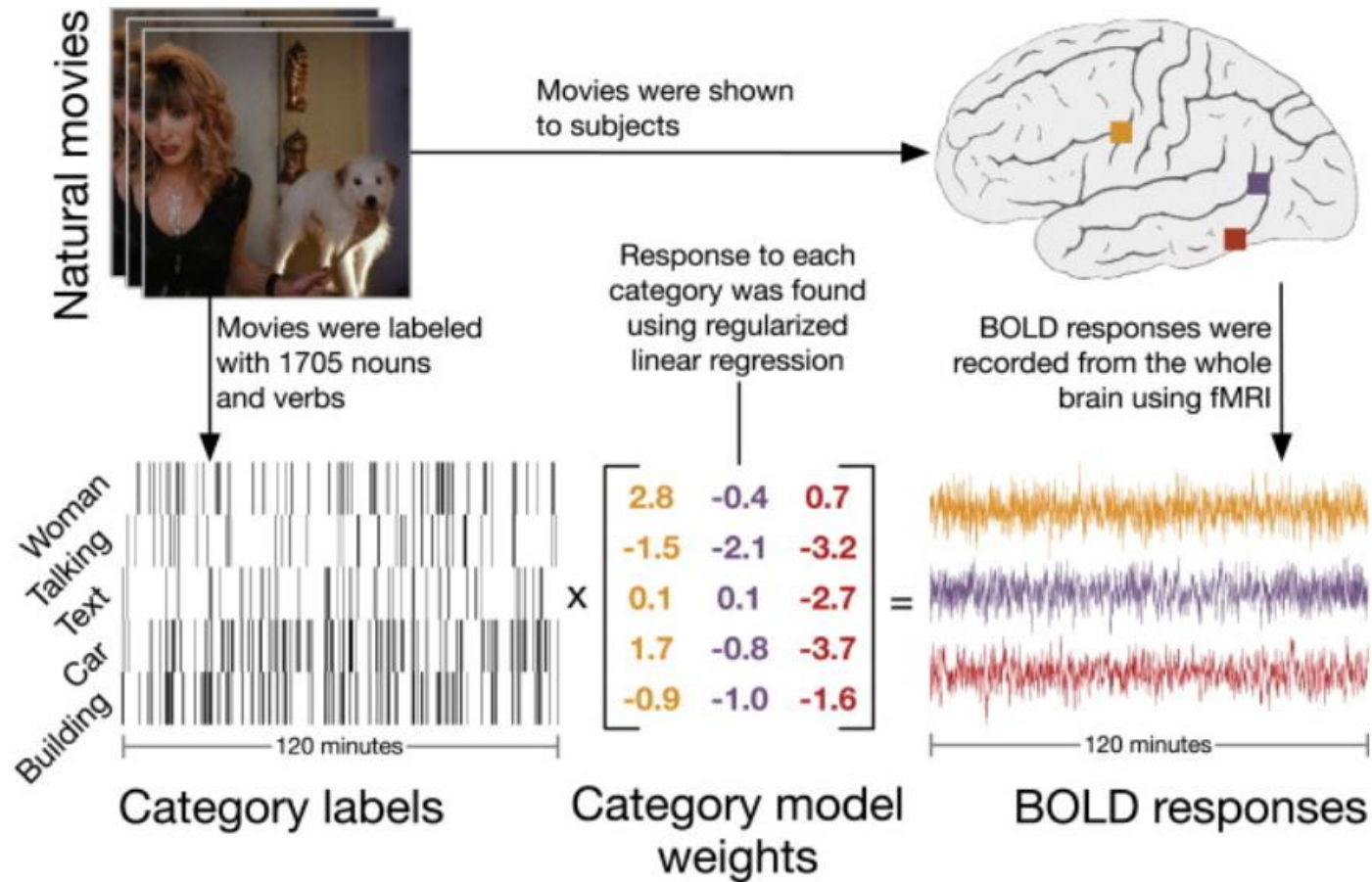
Articulations (22)

Semantic features (985)

Time (s)

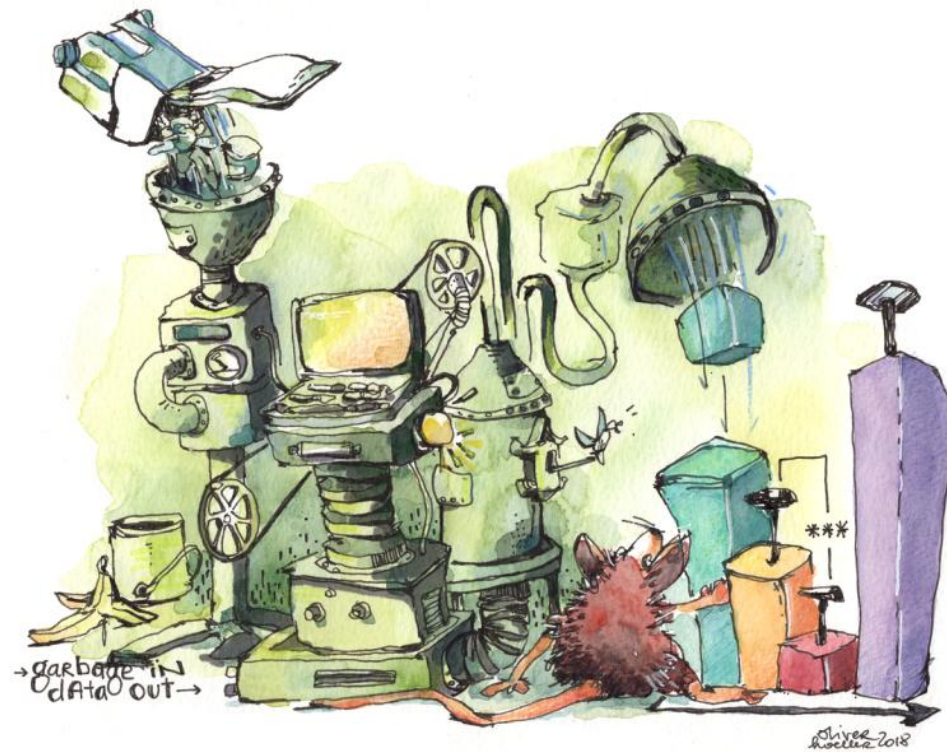
$$\vec{\Phi}(s(t))$$

Computational models

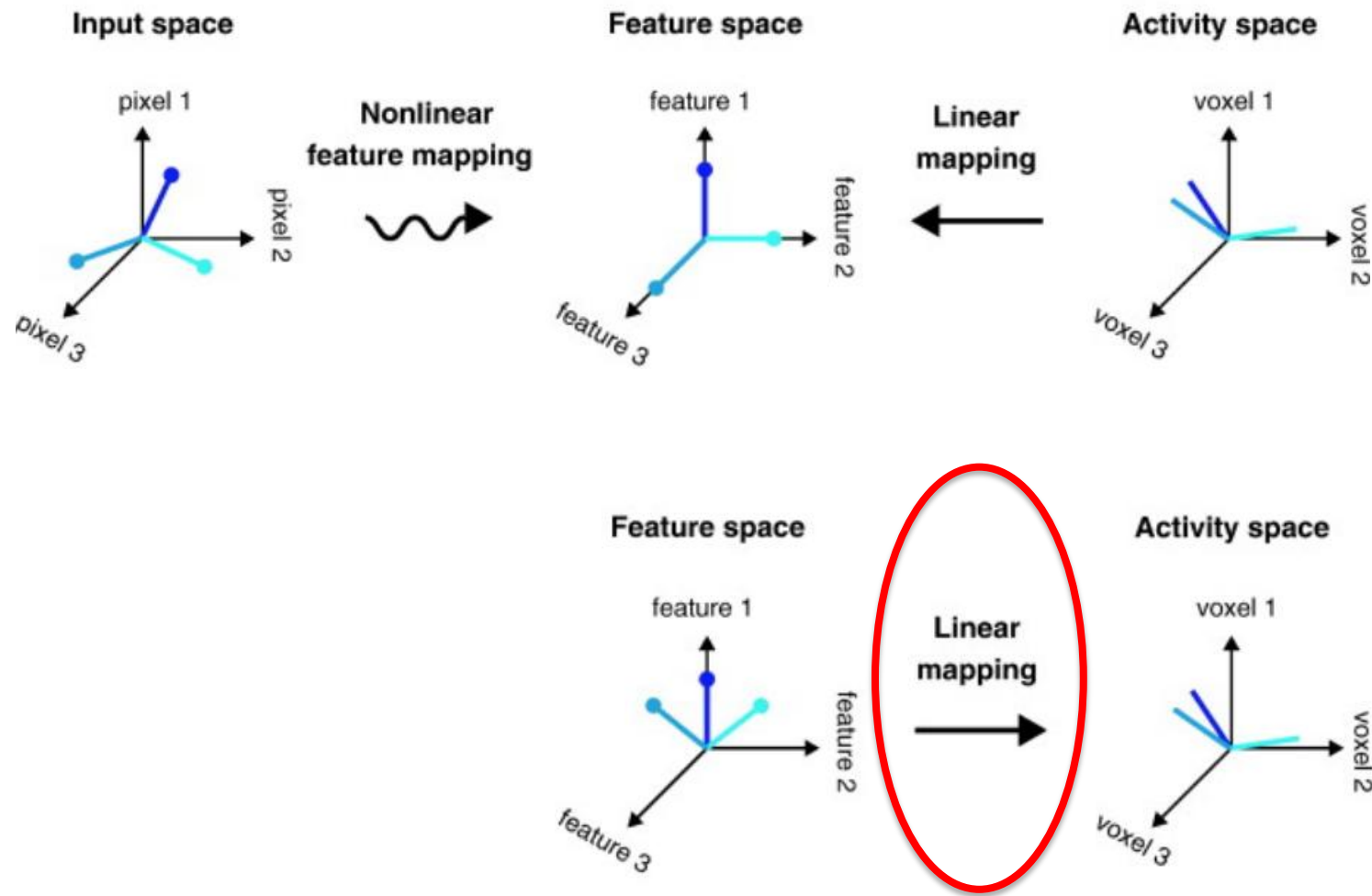


The perils of computational models

- Always check the quality of your computational models (e.g., external or custom-made benchmarks)
- Never forget the phrase: *garbage in, garbage out*



Encoding approach



DECODING

ENCODING

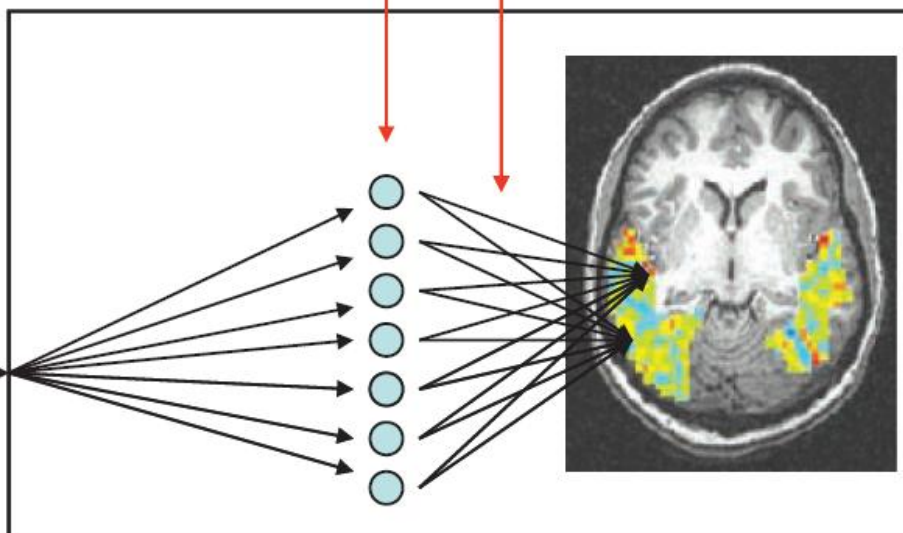
- Encoding is about prediction of brain activity!

A model produces a
feature-based
description of *the
stimulus*

Machine learning
technique to predict
brain activity →

$$Y = \beta X + \varepsilon$$

estimated weights (pointing to β)
 error (pointing to ε)
 brain activity (pointing to Y)
 model features (pointing to X)



Predicted neural
activity for the
stimulus

To predict brain activity, one can:

- perform one procedure only (e.g., a GLM) to predict all the available data in one shot
- learn the Beta weights using a partition of the dataset, and test the goodness of the procedure with the left out data (e.g., cross-validation)
- learn the Beta weights using a dataset, and test the goodness trying to predict brain activity on a completely independent dataset

The latter approach provides an unbiased estimation of the effect size and provides “generalizability”

- Encoding is generally voxel-wise, assuming high granularity of the features (e.g., a feature is not encoded in the pattern!)
- Multivariate encoding (i.e., predict brain activity across multiple voxels simultaneously) is rarely performed (e.g., Canonical Correlation Analysis)

- Encoding approach has the ability to take back brain activity to specific models of interest
- Encoding is still voxel-wise: pay attention with high-level descriptors!
- Significant encoding performance of a single model (at whatever level of performance) does not provide strong constraints for a theory
- Many models must be tested and inferentially compared to drive theoretical progress

